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SUPERSEDING
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(See 6.9)

MILITARY SPECIFICATION

FREQUENCY CHANGER, SOLID STATE (NAVAL SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solid state frequency changers for the conversion of electrical power from 60 hertz (Hz) 3-phase to 400 Hz 3-phase, for Naval surface ship use.

1.2 Classification. The solid state frequency changers shall be of the following types, as specified (see 6.2.1):

Type I - Water-cooled frequency changers, cooled through the use of water-to-air heat exchangers.
Type II - Air-cooled frequency changers, cooled by the ambient air.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6130

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SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915/30 - Cable, Electrical, 1000 Volts, Type TSGU.
MIL-C-915/32 - Cable, Electrical, 1000 Volts, Type MSCU (Including Variations MSCA and MSCS).
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-1310 - Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current (Metric).

2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSEA 314-5724731 - Frequency Changer, Solid State, 250 kW.
NAVSEA 314-5724732 - Frequency Changer, Solid State, 160 kW.
NAVSEA 314-5724734 - Frequency Changer, Solid State, 63 kW.
NAVSEA 314-5724735 - Frequency Changer, Solid State, 285 kW, Air-Cooled.
NAVSEA 314-5724736 - Frequency Changer, Solid State, 180 kW, Air-Cooled.
NAVSEA 314-5724738 - Frequency Changer, Solid State, 72 kW, Air-Cooled.

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(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), prior to production a sample of a solid state frequency changer (hereinafter referred to as equipment) shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Performance requirements.

3.2.1 Input power. Input power to the equipment (see 6.4.1) shall be 440 volts (V), 60 Hz, 3-phase, 3-wire, ungrounded, in accordance with type I power of DOD-STD-1399, section 300. Equipment shall not be damaged when the input voltage phase rotation is in a reverse order (AC-CB-BA).

3.2.1.1 Self-protection from input voltage variation and transient. When the equipment is supplying loads as specified herein, any changes of input voltage or frequency, or both, outside the transient limits as specified in DOD-STD-1399, section 300 and herein, shall not cause damage or fuse blowing in the equipment. Upon return of input voltage and frequency to the specified range (see 3.2.1), the equipment output steady-state voltage and frequency shall be available in less than 1 second and shall operate and supply the load at the same output voltage and frequency prior to the input power changes, without manual restart. The performance of the equipment shall be as specified herein, without any adjustment or replacement of parts. When the input voltage and frequency stay outside the transient limits for more than the time limits specified in table I, the equipment may automatically shut down.

TABLE I. Time limits for transient limits.

Percent of rated load	Time in milliseconds (ms)
20	30
40	15
60	8
80	5

When the input voltage and frequency stay outside the transient limits for more than 70 ms, the equipment shall automatically shut down. Normal starting procedures (see 3.2.1.5) shall be required to restart the equipment. The input voltage and frequency changes outside the transient limit may be any of the following:

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(a) The input voltage instantly drops to zero.
(b) The input voltage drops between 0 and 82 percent of nominal, and input frequency drops between 0 and 92 percent of nominal. The voltage-frequency relation during the period shall be such that voltage will be maintained at nominal until the frequency drops down to about 83 percent of nominal, then the voltage drops in proportion to at a frequency less than 83 percent, and finally the input voltage collapses to zero at a frequency of about 35 percent.
(c) The input voltage rises to 150 percent of nominal, or input frequency rises to 125 percent of nominal, or both.

3.2.1.2 Self-protection from input spikes. When the equipment input voltage experiences a transient (line-to-line or line-to-ground, or both) to 2500 V having a basic impulse level (BIL) wave shape as shown on figure 1, the equipment output voltage waveform shall be as specified in 3.2.2.8. Equipment shall continue to operate without shut down or malfunction. The calculated 60 Hz source surge impedance shall be as specified in table II.

TABLE II. Calculated 60 Hz source surge impedance.

Navy type cable	Current rating of cable (amperes at 50°C)	Surge impedance (ohms)
T14	47	26
T23	64	26
T50	101	20
T75	136	20
T100	160	19
T150	216	18
T200	250	18
T300	320	18
T400	400	17

3.2.1.3 Input current harmonics. When the equipment is powered by a 60 Hz sinusoidal voltage of the specified range (see 3.2.1) and with the equipment supplying any load from no-load to rated-load, the operation of the equipment shall not result in input current waveform distortion as follows:

(a) The total harmonic current shall be not more than 5 percent of the rated-load input current.
(b) No single harmonic current shall be more than as shown on figure 2.

3.2.1.4 Input current balance. When the equipment is supplying any load from 10 percent to rated-load and input 3-phase voltages are balanced within 4 V, the input current shall be balanced to the extent that current in any input line shall not differ by more than 5 percent of the rated-load input current from the arithmetic average current of the three input lines. When the equipment is supplying any load from 10 percent to rated-load and input 3-phase voltages are unbalanced by 3 percent as specified in DOD-STD-1399, section 300, the input current shall be balanced to the extent that the current in any input line shall not differ by more than 10 percent of the rated load input current from the arithmetic average current of the three input lines. The 60 Hz source impedance shall be as specified in table III. The 60 Hz source impedance for the 160 kW, 180 kW, 250 kW and 285 kW units shall be extrapolated from table III.

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TABLE III. 60 Hz, maximum power source impedance in ohms.1/

Load power factor	1.0		0.8		0.7		0.6		0.5		0.4	
Load kVA 3-phase	R	X	R	X	R	X	R	X	R	X	R	X
440-V, 3-phase, 60 Hz												
10	0.81	0.058	0.97	0.058	1.1	0.058	1.2	0.077	1.4	0.077	1.5	0.077
20	.42	.058	.51	.058	0.57	.058	0.64	.067	0.70	.067	0.81	.067
30	.30	.058	.36	.058	.39	.058	.44	.058	.49	.071	.54	.071
40	.23	.058	.28	.058	.29	.058	.33	.058	.37	.067	.42	.067
50	.19	.058	.23	.058	.23	.058	.27	.058	.29	.058	.34	.069
60	.17	.061	.19	.061	.22	.061	.22	.061	.25	.061	.28	.071
70	.15	.060	.17	.060	.18	.069	.20	.069	.22	.069	.24	.069
80	.14	.060	.15	.070	.16	.070	.17	.070	.19	.070	.21	.070
90	.14	.060	.14	.060	.15	.060	.16	.069	.17	.069	.19	.069
100	.12	.060	.13	.060	.14	.069	.15	.069	.16	.069	.17	.069

1/ To convert to per unit values:

$$x(\text{pu}) = \frac{x(\text{ohms})\ (\text{kVA of load})}{(\text{L-L voltage})^2} \times 1000$$

3.2.1.5 Starting time. Equipment shall accept any load from no-load to rated-load and operate as specified herein, in not more than 30 seconds after close of input contactor. Equipment shall not require adjustments, when the input voltage and frequency varies within the range specified in 3.2.1. Means shall be provided within the equipment to automatically preclude the equipment from accepting any connected load during the starting period.

3.2.1.5.1 Prohibition of start. Equipment shall start when the input voltage is within the range of 409 to 471 V and the input frequency is within the range of 57 to 63 Hz. Equipment shall not start when the input voltage is less than 390 V or higher than 490 V, or the input frequency is less than 55 Hz or higher than 65 Hz.

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3.2.1.6 Starting input current. Input current shall be limited during the starting time (see 3.2.1.5). The peak input starting (or in-rush) current shall be not greater than 150 percent of the peak full-load steady-state input current. An auxiliary contactor may be used as a start-up contactor, provided that the start-up contactor is not required for normal operation.

3.2.1.7 No-load input current. The maximum no-load input current shall be not greater than 16 percent of rated-load input current.

3.2.1.8 Grounding. Equipment shall operate as specified herein, when connected to an ungrounded ship's input power system having 3 to 20 microfarads capacitance from any line-to-ground, or with any single input or output line or terminal intermittently or permanently grounded. Equipment which requires grounding, either directly or by means of capacitors, shall be isolated from shipboard electrical power systems (both 60 Hz and 400 Hz) through transformers if leakage current exceeds 30 milliamperes per phase. Maximum filter capacitance shall not exceed 0.1 microfarad for 60 Hz output and 0.02 microfarad for 400 Hz output.

3.2.1.9 Input power factor (p.f.). The input power factor presented to the 60 Hz input power by the equipment, shall be between 0.75 lagging and 0.9 leading.

3.2.2 Output voltage. Rated output voltage shall be 450 V root mean square (Vrms). The output circuit shall be delta-connected, ungrounded, 3-phase, 3-wire, 400 Hz for all units. The phase rotation of the output voltage shall be a clockwise sequence of AB-BC-CA.

3.2.2.1 Voltage regulation. The average of the Vrms values of the output voltages shall be maintained at the regulated point (at the remote sensing point or at the equipment output terminals (see 3.2.2.2)) within plus or minus 0.5 percent of rated output voltage, under the following variable conditions and any combination thereof, at all voltage settings within the adjustment range specified in 3.2.2.3:

(a) A voltage drop from the terminals of the equipment to the regulated point of any amount from 0 to plus 7 percent of rated output voltage, as specfied in 3.2.2.2.
(b) A load variation from no-load to 125 percent of rated-load.
(c) Load p.f. varying between 0.6 lagging and 0.9 leading.
(d) An input voltage and frequency variation within the range specified herein.
(e) Variation of the equipment temperature from cold start (at ambient temperature ranging from 0 to 50 degrees Celsius (°C)) to maximum operating temperature (at ambient temperature ranging from 0 to 50°C).
(f) Under the unbalanced load conditions specified in 3.2.2.11.
(g) Under the pulsing nonlinear load conditions specified in 3.2.2.1.1.
(h) Under the steady-state nonlinear load conditions specified in 3.2.2.1.2.

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3.2.2.1.1 Pulsing nonlinear load. Power factor of any pulsing loads shall be between 0.9 lagging and unity. The pulsing loads shall be nonlinear loads and have a synchronous variable amplitude with dominant low frequency mode between 1 and 50 Hz and with a rise time of 40 ms. When a pulsing load of 25 percent of rated-load is applied to the equipment, the equipment output voltage regulation shall be in accordance with 3.2.2.1 and the equipment output voltage modulation shall be in accordance with 3.2.2.7.

3.2.2.1.2 Steady-state nonlinear loads. Typical steady-state nonlinear loads shall be defined as loads developed by the 3-phase full-wave rectifying systems, and shall have maximum harmonic current content imposed on the equipment output line in accordance with the following formula, when the load system is supplying resistive loads:

$$I_N = \frac{100}{N} \text{ percent of fundamental current}$$

where I_N is the Nth harmonic current and N is the harmonic number. N shall not be any even number or multiple of three. When a nonlinear load up to 70 percent of the rated-load is applied to the equipment, the output voltage waveform shall remain within the limits as specified in 3.2.2.8, the output voltage regulation shall be in accordance with 3.2.2.1, and the voltage amplitude modulation shall be in accordance with 3.2.2.7.

3.2.2.2 Output voltage remote sensing. Equipment shall operate with output voltage sensing at a remote point. The remote voltage sensing point will be connected to the unit by a type TSGU-3 cable in accordance with MIL-C-915/30, or equivalent cable. The cable will be furnished by the installation activity. The maximum length of the remote sensing cable shall be 200 feet. For any loads between no-load and 125 percent of rated-load (see 3.2.6), the voltage at the remote sensing point shall be automatically maintained at the adjusted voltage (see 3.2.2.3) and shall conform to all steady-state voltage regulation, amplitude modulation, waveform, phase displacement, balance, unbalance, pulsing loads and nonlinear loads as specified (see 3.2.2.1, 3.2.2.7, 3.2.2.8, 3.2.2.9, 3.2.2.10, 3.2.2.11, 3.2.2.1.1, and 3.2.2.1.2, respectively). The voltage drop between the terminals of the unit and the remote voltage sensing point shall not exceed 7 percent of rated voltage at 125 percent of rated-load. In the event that the remote voltage sensing signal is disconnected or otherwise interrupted, the equipment shall automatically shift the voltage sensing from the remote point to the unit output terminals. Under these conditions the voltage at the unit terminals shall be within plus or minus 2 percent of the adjusted voltage level, which existed at the remote point prior to transfer. An indication light shall be provided on the remote monitoring panel to show the loss of the remote voltage sensing signal. After restoration of the remote voltage sensing, the voltage regulation shall automatically shift back to the remote sensing point.

3.2.2.3 Voltage adjustment. Means shall be provided to adjust the output voltage at the regulated point over a range of plus or minus 5 percent of the rated voltage, with adjustment resolution not more than 150 millivolts (mV) at any load from no-load to 125 percent rated-load under any steady-state input conditions. The adjustment device shall have a slotted shaft for screwdriver type adjustment. Test points shall be provided for the connection of an external voltmeter which will indicate the unit output voltages.

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3.2.2.4 Transient output voltage and recovery. The sudden change of 0.8 lagging p.f. loads on the equipment from no-load to half-load, from half-load to rated-load, from rated-load to half-load and from three-fourths load to one-fourth load shall cause an output voltage deviation of not more than plus or minus 5.0 percent of rated voltage. The voltage at the equipment output terminals shall recover and stay within the steady-state regulation band (see 3.2.2.1) within 0.02 second. The sudden change of 0.8 lagging p.f. loads on the equipment from no-load to rated-load and from rated-load to 5 percent load shall cause an output voltage deviation of not more than plus or minus 10 percent of rated voltage. The output voltage shall recover and stay within the steady-state regulation band within 0.02 second. Circuits shall be designed so that voltage transients developed within the equipment shall not reflect back into the input power system such that transient limits shall not exceed alternating current (a.c.) input in accordance with type I power of DOD-STD-1399, section 300.

3.2.2.4.1 Transient output voltage and recovery for electronic transient loads. Electronic transient loads shall be defined as the sudden turn-ons of an electronic type load such as radar high voltage direct current (d.c.) power supplies or the sudden turn-offs of a radar system by its protective device (crowbar), whose p.f. is approximately 0.6 lagging. Rise time of the electronic transient load from 0 to the maximum current shall be not less than 5 ms. When an electronic transient load up to 125 percent of the rated unit load is applied, the equipment output voltage deviation shall be not more than plus or minus 13 percent of the rated voltage. The voltage at the equipment output terminals shall recover and stay within the steady-state regulation band (see 3.2.2.1), within 20 ms after application of the electronic transient load.

3.2.2.5 Frequency regulation. Output frequency shall be maintained within plus or minus 0.5 percent of rated frequency under the following conditions and combination thereof:

(a) A load variation from no-load to rated-load.
(b) Load p.f. varying between 0.6 lagging and 0.9 leading.
(c) An input voltage and frequency variation within the range specified herein.
(d) Variation of the equipment temperature from cold start (at ambient temperature ranging from 0 to 50°C) to maximum operating temperature (at ambient temperature ranging from 0 to 50°C).
(e) Variation of output voltage specified in 3.2.2.3.

3.2.2.6 Transient frequency deviation and recovery. A sudden change of 0.8 lagging p.f. loads on the equipment from no-load to rated-load and from rated-load to no-load, shall not cause a frequency deviation greater than the steady-state band specified in 3.2.2.5.

3.2.2.7 Amplitude and frequency modulation. Amplitude and frequency modulation of the output voltage, as specified in DOD-STD-1399, section 300, shall not exceed 1 percent and 0.5 percent, respectively, under any of the conditions or combinations of conditions specified in 3.2.2.5.

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3.2.2.8 Output voltage waveform. Under any load conditions between no-load to rated-load, the rms value of the sum of all of the harmonic content in the output voltage shall not exceed 3 percent of the rated output voltage. The rms value of any single harmonic shall not exceed 2 percent of the rms value of the rated output voltage. The deviation factor, as specified in DOD-STD-1399, section 300, of the output voltage waveform shall not exceed 5 percent. Peak-to-peak amplitude of voltage spikes (between the phases) of any frequencies shall not exceed 3 percent of the rms value of the rated output voltage.

3.2.2.9 Output phase displacement. Displacement angle between adjacent voltage phases shall be 120 $\pm$ 1 degrees under all conditions of operation specified herein.

3.2.2.10 Output voltage balance under balanced load conditions. The difference between phase voltages at the voltage regulated point of the equipment shall be not greater than 0.5 percent of rated voltage at no-load and at any balanced 3-phase load up to rated current with any p.f. between 0.6 lagging and 0.9 leading.

3.2.2.11 Output voltage balance under unbalanced load conditions. The voltage variation between phases at the equipment output terminals shall be not greater than 1 percent of the rated voltage under the unbalanced load conditions shown in table IV, when the output voltage is regulated at the equipment output terminals. Equipment shall operate with a single phase 0.6 p.f. load carrying a current 7 percent of rated current. The voltage of the loaded phase at the voltage regulated point shall have regulation as specified in 3.2.2.1.

TABLE IV. Unbalanced loads.

Load conditions	Line A (percent of rated current)	Line B (percent of rated current)	Line C (percent of rated current)	Power factor
1	87	80	80	0.6 lagging to unity
2	80	87	80	0.6 lagging to unity
3	80	80	87	0.6 lagging to unity

3.2.2.12 Input voltage change effect. When the equipment is supplying any load from no-load to rated-load and with any output and input steady-state voltage and frequency within the range specified in 3.2.1, the occurrence of any input transient voltage up to 16 percent, with recovery time of 2 seconds as specified for type I power of DOD-STD-1399, section 300, shall not cause an output voltage deviation of not more than plus or minus 3 percent of the initial value. The output voltage shall recover to and remain within the steady-state regulation band, within 0.02 second from the initiation of the input voltage change.

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3.2.3 Power ratings. The power ratings of the equipment shall be 63, 72, 160, 180, 250 and 285 kW as specified (see 6.2.1) at 0.8 lagging p.f. The kilovoltampere (kVA) rating and p.f. rating for each of the corresponding kW rating shall be within the power range (shaded area) as shown on figure 3.

3.2.3.1 Duty. Equipment shall operate continuously at the specified power rating within the p.f. range as shown on figure 3.

3.2.3.2 Motor starting. Equipment shall start a 400 Hz motor having starting current up to 125 percent of the rated current. The motor starting p.f. shall be 0.3 lagging and the starting current shall last for 30 seconds. During the motor starting, for that portion of load cycle when the p.f. of the total load is between 0.8 lagging and unity, the output transient voltage of the unit shall be as specified in 3.2.2.4.

3.2.4 Efficiency. The minimum overall efficiency of the equipment, when operated at the rated input voltage and frequency (see 3.2.1), shall be as specified in table V.

TABLE V. Efficiency.

Unit rating in kW		Percent efficiency at 50 percent load, 0.8 to 1.0 p.f.	Percent efficiency at 100 percent load, 0.8 to 1.0 p.f.
Water cooled	Air cooled		
63	72	79	84
160	180	81	84
250	285	82	84

3.2.5 No-load losses. No-load losses of the equipment shall be not more than 80 percent of the rated load losses at nominal input voltage and frequency.

3.2.6 Overload. Equipment shall provide and sustain an overload current of 125 percent of rated current at the rated output voltage for 1 minute and meet all requirements specified herein (see figure 4). When the overload current (between 100 and 125 percent of rated-load) lasts 1 minute $\pm$ 5 seconds, the equipment shall automatically shut down. An indicator shall be provided to show overload shutdown. The output voltage regulation for any overload current up to 125 percent of the rated output current shall be as specified in 3.2.2.1. The equipment shall also provide and sustain an overload current of more than 125 percent and up to 150 percent of rated current for 1 minute $\pm$ 5 seconds. When the overload current is between 125 percent and up to 150 percent of rated current, the equipment shall maintain the rated output voltage for a minimum of 1 second. Thereafter, the equipment output voltage may decrease and operate as specified in 3.2.7.2 and 3.2.7.3.

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3.2.7 Fault and malfunction protection.

3.2.7.1 Fault isolation. Equipment shall incorporate fault isolation for each module or drawer (see 6.4.4 and 6.4.5) to prevent cascading to other modules or drawers. A shorted or noncommutated power semiconductor shall not result in damage to any other part (see 6.4.6) of the equipment, except that a single fuse may blow.

3.2.7.2 Fault current. Output current between 125 and 135 percent of rated current is considered fault current. Equipment shall provide and sustain any fault current up to 135 percent of rated current for 1 minute ± 5 seconds when a fault develops at any external circuits supplied by the equipment. After the equipment has supplied the fault current for 1 minute ± 5 seconds, the equipment shall automatically shut down. When the fault current is between 125 and 135 percent of rated current, the undervoltage protection (see 3.2.7.4) shall not shut down the equipment. An indicator shall be provided to show fault current and short circuit shutdown.

3.2.7.3. Short circuit protection. Output current above 135 percent of rated current is considered short circuit current. Equipment shall protect itself without damage to any parts from any short circuit current on any or all output phases. Protection shall be provided by current limiting to a value not less than 135 percent or greater than 160 percent of rated current (see figure 4). The equipment shall provide and sustain the short circuit current for 1 minute ± 5 seconds. After the equipment has supplied the short circuit current for 1 minute ± 5 seconds, the equipment shall automatically shut down. When the short circuit current is equal to or less than 150 percent, the equipment shall maintain the current for a minimum of 1 second and then the current may be changed into the current limiting value. When the initial short circuit current is greater than 150 percent, the current may be reduced to the current limiting value in less than 1 second. The current limiting shall not be applied to any output current less than 135 percent of rated current. On failure of the current limiting circuits, the equipment shall be protected by fuses.

3.2.7.4 Protection for utilization equipment. A protective scheme shall be provided to shut down the equipment, if the output voltage at the regulated point or frequency conditions or both as shown in table VI exists. The equipment shutdown shall be delayed from a minimum of 45 ms to a maximum of 250 ms to prevent nuisance tripping due to possible system transients. The undervoltage protection shall be inhibited under the fault condition, as specified in 3.2.7.2.

TABLE VI. Shutdown voltage and frequency settings.

Conditions	Settings (percent of rated output)	Tolerance (percent of rated output)
Overvoltage	115	plus or minus 5
Undervoltage	85	plus or minus 5
Overfrequency	105	plus or minus 1
Underfrequency	95	plus or minus 1

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3.2.8 Parallel operation. Equipment shall operate in parallel, at the levels of reliability and maintainability specified herein, to a maximum of six units without interrupting the load and without introducing transient voltages at the input line and output load bus outside the specified limits. After the units are connected for parallel operation and the output voltage of each unit has been manually adjusted (see 3.2.2.3) so that the maximum terminal voltage difference between the units is not more than 1 V, the synchronization and parallel operation of the units shall be automatically accomplished to meet all requirements specified herein. The units shall operate in parallel under any load conditions from no-load to rated-load.

3.2.8.1 Synchronization. In parallel operation, an oscillator from any one of the six units shall be automatically selected as the master driving or synchronizing unit for the oscillators in the other five units. The frequency of the master oscillator shall be continuously monitored to enable one of the other five oscillators to be automatically substituted, whenever the master oscillator drifts out of tolerance or malfunctions.

3.2.8.2 Real and reactive load division. When paralleling identical size units, the maximum difference in kW supplied by the units in parallel shall not exceed 10 percent of the kW rating of one of the units under load conditions between 10 and 100 percent of the total rated-load of the paralleled units. The maximum difference in kilovar (kvar) supplied by the units operating in parallel shall not exceed 10 percent of the kvar rating of one of the units under any load condition between 10 and 100 percent of the total rated-load of the paralleled units. When paralleling different size units, the output of each unit shall be directly proportioned to the kW rating of the individual unit. The maximum difference in kW between the actual kW supplied by each unit and its proportional share of the total kW, shall not exceed plus or minus 5 percent of the average kW rating of units in parallel under load conditions between 10 and 100 percent of the total rated-load of the paralleled units. The maximum difference in kvar between the actual kvar supplied by each unit and its proportional share of the total kvar shall not exceed plus or minus 5 percent of the average kvar rating of units in parallel under load conditions between 10 and 100 percent of the total rated-load of the paralleled units. The above requirements shall apply for input voltage variations specified in 3.2.1 and other environmental requirements specified herein.

3.2.8.3 Selective tripping. Equipment shall operate so that when any malfunction occurs to any one of the units operated in parallel, the malfunctioned unit shall automatically shut down and the remaining units shall continue to operate in parallel, as specified herein.

3.2.8.4 Load transfer to another source. The equipment shall operate in parallel with another source (such as motor-generators) having the same output voltage and frequency for the purpose of transferring any load up to rated-load from the equipment to another source, or vice versa. The paralleling for load transfer shall be achieved without power interruption to the load. The equipment shall be protected from damage in the event that the paralleling is inadvertently attempted when the equipment and the other source are out of phase by 10 electrical degrees.

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3.2.8.5 Arrangement of parallel operation. The parallel operation of up to the maximum six units shall be connected to a common bus (see 3.2.8.5.1) or to a split bus (see 3.2.8.5.2) arrangement. Paralleling, as specified in 3.2.8, 3.2.8.1, 3.2.8.2, 3.2.8.3 and 3.2.8.4, shall apply to both the common bus and the split bus arrangements.

3.2.8.5.1 Common bus parallel operation. The common bus parallel operation shall be connected as shown on figure 5. When the 400 Hz bus is located in the same compartment with the frequency changers such that the impedance of the 400 Hz cable between the units and the 400 Hz bus shall be negligible, the voltage of the common bus shall meet all output requirements herein (see 3.2.2 through 3.2.2.12).

3.2.8.5.2 Split bus parallel operation. The split bus parallel operation shall be connected as shown on figure 6 up to the maximum of six units and three buses in any combination. When each unit is located close to its own 400 Hz bus such that the impedance of the 400 Hz cable between the unit and its bus shall be considered negligible and the maximum impedance of each bus tie cable is less than 2 percent (based on the rated output current of one unit), the voltage at one of the 400 Hz buses as A, B or C (bus C is not shown on figure 6) shall meet all output requirements as specified (see 3.2.2 through 3.2.2.12). Selection of the bus to meet the output voltage requirements may be accomplished by a control cable connecting between the unit and its bus. One normally-open and one normally-closed auxiliary contact are available in each bus tie breaker and can be used to provide bus tie breaker status information, for control of the split bus parallel operation.

3.3 Design and construction.

3.3.1 Basic requirements. In addition to the requirements of this specification, the equipment shall be in accordance with MIL-E-917 except the following paragraphs and subparagraphs pertaining thereto.

Fundamental equipment characteristics (excluding safety)
Grounding
Input power variations (alternating current equipment)
Inclined operation
Ambient temperature
Impact shock and vibration
Equipment mounting
Radio interference
Noise reduction for submarine
Piping system
Drawings
Manuals
Workmanship.

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3.3.2 Construction.

3.3.2.1 Fabrication and construction. Each unit shall be fabricated and constructed in accordance with the following, as applicable:

(a) Drawing 314-5724731 for 250 kW water-cooled unit.
(b) Drawing 314-5724732 for 160 kW water-cooled unit.
(c) Drawing 314-5724734 for 63 kW water-cooled unit.
(d) Drawing 314-5724735 for 285 kW air-cooled unit.
(e) Drawing 314-5724736 for 180 kW air-cooled unit.
(f) Drawing 314-5724738 for 72 kW air-cooled unit.

Mating electrical connectors shall be as specified (see 6.2.1).

3.3.2.2 Dielectric withstanding voltage and insulation resistance. Equipment, except the control drawer, shall pass insulation resistance and dielectric withstanding voltage tests specified in 4.7.1 and 4.7.2. Insulation resistance of the circuits shall be not less than 10 megohms when measured with units at ambient temperature of 25 ± 2°C. This requirement shall be met without resorting to either disassembly or disconnection of any unit, cabinet, drawer (with the exception of the control drawer), module or part. There shall be no interconnection of circuits, shorting together of terminals, short circuiting, by-passing or grounding of parts or circuit elements during the test to prevent the test voltage from being applied to parts. Such measures, however, may be employed when specifically authorized in the accepted test procedure for convenience and effectiveness in testing the equipment, provided the measures employed do not prevent the test from being effective on portions of the equipment. Use of any such measures in the test procedure shall be entirely at the option of the acquisition agency based upon recommendations and justification to be provided by the contractor. The design of the equipment shall be such that required insulation resistance and dielectric withstanding voltage tests on the insulation system can be conducted without either incipient or catastrophic damage or without significant (within acceptable product variability tolerance) degradation to any material or part, including the electrical insulation. Furthermore, a defect in the insulation system shall not cause either of the tests, when properly conducted, to result in damage or degradation to any other material or part except that which may be inflicted to the immediate adjacent parts separated by the defective insulation.

3.3.2.3 Size and weight. The equipment shall not exceed the weight and dimensions shown in table VII.

TABLE VII. Dimension, weight and number of cabinets.

Rating kW		Number of cabinets	Number of cabinets in a section	Number of sections in one unit	Dimensions of each section			Maximum weight
Water cooled	Air cooled				width (inches)	depth (inches)	height (inches)	per unit (lb)
63	72	3	3	1	72	27	80	5,300
160	180	6	3	2	72	27	80	11,500
250	285	8	4	2	96	27	80	14,500

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3.3.3 Remote control. Signals for the remote control, indication and interlock shall be provided on a remote control panel for single unit and parallel operation. The remote control panel shall be connected to the unit by a type MSCU cable in accordance with MIL-C-915/32 or equivalent. The cable will be furnished by the installation activity. The maximum length of the interconnection cable between the equipment and the remote control panel shall be 500 feet. The controls and indicators on the remote control panel shall be as follows (the remote control panel and its associated controls and indications will be contracted by the installation activity; all controls and indications shown on figures 5 and 6 are on the remote control panel):

(a) Pushbutton or switch for unit startup and shutdown.
(b) Pushbutton or switch for output contactor close and open.
(c) Indication light to show that the unit is energized.
(d) Indication light to show that the output contactor is closed.
(e) Output current ammeter.
(f) Parallel operation indication light to show that the units are in synchronization and parallel operation interlock (normally open contact) to permit the closing of bus tie circuit breakers.
(g) Indication light to show that the unit is in remote control.

3.4 Environmental requirements.

3.4.1 Shock. Equipment shall withstand the high-impact, grade A, class III, type A shock test of MIL-S-901 for deck mounted equipment without the use of shock isolators on the base of the equipment. Resilient mounts may be used to reduce the airborne and structureborne noise levels of the equipment. If resilient mounts are used for noise attenuation, they shall be tested with the equipment.

3.4.2 Vibration. Equipment shall be designed and tested in accordance with type I vibration of MIL-STD-167-1 up to and including 33 Hz. Equipment shall withstand the vibration test without any sway braces.

3.4.3 Noise. Equipment shall be either solidly mounted or resiliently mounted to the ship structure and shall be designed and tested in accordance with MIL-STD-740, except that the airborne noise requirements shall be as specified in table VIII and structure-borne noise requirements shall be as specified on figure 7.

TABLE VIII. Airborne noise acceptable sound pressure levels in dB re 20 µPa.1/

Octave band center frequency in Hz	31.5	63	125	250	500	1K	2K	4K	8K
Sound pressure level1/ (dB re 20 µPa)	95	90	85	83	87	88	85	76	75

1/ Decibels (dB) relative to 20 micropascals (µPa).

The airborne and structureborne noise levels specified herein shall be for the complete unit (see 6.4.2) or for each section of a unit when more than one section is in a unit.

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3.4.4 Electromagnetic interference (EMI) emission and susceptibility. Equipment shall conform to the electromagnetic interference limits in accordance with part 5 of MIL-STD-461 for power supplies. Equipment shall conform to only the requirements of CE01, CE03 and RE02. Cable and cabinet bonding and grounding shall conform to MIL-STD-1310. The equipment shall be designed to minimize susceptibility of its control circuits to EMI in accordance with MIL-STD-461.

3.4.5 Ambient temperature. Equipment shall start and operate as specified herein throughout an ambient temperature range of 0 to 50°C.

3.4.6 Cooling.

3.4.6.1 Cooling water. The cooling water for the water-cooled units (see 1.2) shall be sea water within a temperature range of minus 2 to plus 35°C and within a pressure range of 20 to 180 pounds per square inch (lb/in^2).

3.4.6.2 Cooling air. The cooling air for the air-cooled units (see 1.2) shall be ambient air of a maximum temperature of 50°C.

3.4.6.3 Cooling system failure protection. Protection of the equipment against cooling fan failure or cooling water failure shall be provided by the automatic de-energizing of the equipment.

3.4.7 Heat dissipation to ambient. The heat dissipation to the ambient for the equipment (see 1.2) shall be not more than the maximum values specified in table IX, when the ambient temperature is in the range of 25 to 50°C.

TABLE IX. Heat dissipation to ambient.

Unit rating (kW)	Maximum heat dissipation (kW)
63	3
160	4
250	6
72	14
180	34
285	54

3.4.8 Humidity conditions. Equipment shall operate with relative humidities ranging to 95 ± 5 percent by both intermittent and continuous periods, including conditions where condensation takes place in and on the equipment cabinets, in the form of both water and frost.

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3.4.9 Clearance for ventilation and maintenance. Clearance required between the equipment and adjacent equipment or structure shall be not greater than as specified in table X.

TABLE X. Clearance for ventilation and maintenance.

	Water-cooled unit	Air-cooled unit
Bottom	none	none
Sides or ends	3 inches	3 inches
Rear	4 inches	8 inches
Top	8 inches	8 inches
Front	30 inches	30 inches

3.5 Reliability. The reliability measure shall be mean-time-between-failure (MTBF) in accordance with MIL-STD-781 and shall be not less than 4000 hours.

3.5.1 Definition of failure. The definition of failure for the purpose of determining the achievement of reliability characteristics shall be:

(a) The input and output performance not as specified herein.
(b) Malfunction requiring shutdown because of damage or incipient damage.

3.6 Workmanship. Equipment, including parts and accessories, shall exhibit neatness and thoroughness of soldering, marking of parts and subassemblies, wiring, welding and brazing, plating, riveting, finishes, machine operations, screw assemblies and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Calibration system requirements. The contractor shall provide and maintain a calibration system for test equipment in accordance with the data ordering document (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

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4.3 Test conditions. Unless otherwise specified herein, tests shall be conducted under ambient temperature 25 ± 2°C. The specific ambient temperature at the time of the tests shall be measured and recorded. Transient voltage or current traces taken by oscillograph or pictures of oscilloscope shall have peak-to-peak magnitude of not less than 2.5 inches and the speed of the traces shall be not greater than 2 cycles per inch. Oscillograph, oscilloscope or its equivalent shall be used for all transient measurements.

4.4 First article inspection. When specified (see 6.2.1), one unit of each power rating shall be subjected to the examination and tests specified in table XI.

TABLE XI. First article inspection.

Examination and tests	Requirement paragraph	Inspection paragraph
General examination	3.3.1 and 3.6	4.6.1 and 4.6.2
Insulation resistance	3.3.2.2	4.7.1
Dielectric withstanding voltage 1/	3.3.2.2	4.7.2
Self-protection from input voltage variation and transient	3.2.1.1	4.7.3
Self-protection from input spikes	3.2.1.2	4.7.4
Input current harmonics	3.2.1.3	4.7.5
Starting current and starting time	3.2.1.5, 3.2.1.5.1 and 3.2.1.6	4.7.6
Voltage and frequency regulation	3.2.2.1 and 3.2.2.5	4.7.7
Output voltage remote sensing	3.2.2.2	4.7.8
Voltage adjustment	3.2.2.3	4.7.9
Transient voltage and frequency deviation and recovery	3.2.2.4 and 3.2.2.6	4.7.10
Amplitude and frequency modulation	3.2.2.7	4.7.11
Output voltage waveform	3.2.2.8	4.7.12
Output phase displacement angle	3.2.2.9	4.7.13
Output voltage balance under unbalanced load	3.2.2.11	4.7.14
Protection for utilization equipment	3.2.7.4	4.7.15
Input voltage change effect	3.2.2.12	4.7.16
Efficiency and no-load input current	3.2.1.7, 3.2.4 and 3.2.5	4.7.17
Power rating	3.2.3 and 3.2.3.1	4.7.18
Input current balance	3.2.1.4	4.7.19
Overload	3.2.6	4.7.20
Short circuit and fault current	3.2.7.2 and 3.2.7.3	4.7.21
Parallel operation	3.2.8, 3.2.8.1, 3.2.8.2 and 3.2.8.3	4.7.22
Output voltage balance under balanced load	3.2.2.10	4.7.23

See footnote at end of table.

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TABLE XI. First article inspection. - Continued

Examination and tests	Requirement paragraph	Inspection paragraph
Shock	3.4.1	4.7.24
Vibration	3.4.2	4.7.25
Noise (structureborne)	3.4.3	4.7.26
Noise (airborne)	3.4.3	4.7.27
Cooling system failure protection	3.4.6.3	4.7.28
Electromagnetic interference emission and susceptibility	3.4.4	4.7.29
Temperature rise and high temperature operation	3.4.5	4.7.30
Cooling	3.4.6, 3.4.6.1 and 3.4.6.2	4.7.30
Grounding	3.2.1.8	4.7.31
Humidity	3.4.8	4.7.32
Remote startup and shutdown	3.3.3	4.7.33
Motor starting	3.2.3.2	4.7.34
Load transfer to another source	3.2.8.4	4.7.35
Reliability 1/	3.5	4.7.36
Low temperature operation	3.4.5	4.7.37
Pulsing load and nonlinear load	3.2.2.1.1 and 3.2.2.1.2	4.7.38
Transient output voltage and recovery for electronic load	3.2.2.4.1	4.7.39
Input power factor	3.2.1.9	4.7.40

1/ Any of these tests may be conducted concurrently except that the dielectric withstanding voltage test shall be conducted after all other tests, but before the reliability tests have been completed. The reliability tests shall be conducted only after satisfactory completion of all other tests. For the high temperature test as specified in 4.7.30.5, the cooling water inlet temperature of the water-cooled unit shall be maintained at an average of 32°C and within the temperature range between 29 and 35°C. For all other tests, the cooling water inlet temperature of the water-cooled unit shall be maintained within the range between minus 2 and 35°C. For any specific test, the inlet water temperature shall vary not more than 6°C. The temperatures of the cooling water inlet and output shall be measured and recorded for each test. Tap water may be used as cooling water for tests.

4.4.1 First article inspection report. A first article inspection report shall be submitted in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.5 Quality conformance inspection. Each unit offered for delivery shall be subjected to the quality conformance inspection specified in table XII.

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TABLE XII. Quality conformance inspection.

Examination and tests	Requirement paragraph	Inspection paragraph
General examination	3.3.1 and 3.6	4.6.1 and 4.6.2
Insulation resistance	3.3.2.2	4.7.1
Self-protection from input voltage variation and transient	3.2.1.1	4.7.3
Input current harmonics	3.2.1.3	4.7.5
Starting current and starting time	3.2.1.5, 3.2.1.5.1 and 3.2.1.6	4.7.6
Voltage and frequency regulation	3.2.2.1 and 3.2.2.5	4.7.7
Voltage remote sensing	3.2.2.2	4.7.8
Voltage adjustment	3.2.2.3	4.7.9
Transient voltage and frequency deviation and recovery	3.2.2.4 and 3.2.2.6	4.7.10
Amplitude and frequency modulation	3.2.2.7	4.7.11
Output voltage waveform	3.2.2.8	4.7.12
Output voltage balance under unbalanced load	3.2.2.11	4.7.14
Input voltage change effect	3.2.2.12	4.7.16
Overload	3.2.6	4.7.20
Short circuit and fault current	3.2.7.2 and 3.2.7.3	4.7.21
Parallel operation	3.2.8, 3.2.8.1, 3.2.8.2 and 3.2.8.3	4.7.22
Output voltage balance under balanced load	3.2.2.10	4.7.23
Noise (structureborne)	3.4.3	4.7.26
Noise (airborne)	3.4.3	4.7.27
Endurance run	3.2.3 and 3.5	4.7.41

1/ Any of these tests specified in table XII may be conducted concurrently except that the endurance run shall be conducted after all other tests have been completed, to verify that all requirements have been met.

4.5.1 Test reports. The contractor shall prepare quality conformance test reports in accordance with the data ordering document (see 6.2.2).

4.6 General examination. Parts on assemblies and completed units shall be examined as specified in 4.6.1 and 4.6.2.

4.6.1 Major parts. Examination shall be conducted to determine that:

(a) The following items conform to the applicable specification and that the size, type and rating are as shown on the applicable drawing specified in 3.3.2.1:

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(1) Resistors and rheostats.
(2) Instruments, instrument transformers, shunts and other associated accessories.
(3) Capacitors.
(4) Switches.
(5) Semiconductor devices.
(6) Terminal boards.
(7) Laminated materials such as glass mica glass (GMG) and glass silicon glass (GSG) used for terminal boards, sub-panels and so forth.

(b) Materials used in the construction of the following wound components shall comply as shown on the applicable drawing specified in 3.3.2.1:

(1) Windings with varnish impregnation in accordance with the procedures shown on the applicable drawings specified in 3.3.2.1.
(2) The resistance of all windings shall be as shown on the drawing specified in 3.3.2.1. Number of turns shall be as shown on the applicable drawing specified in 3.3.2.1.

4.6.2 Completed equipment. Examination of the completed equipment shall be conducted to determine that:

(a) Parts, modules and drawers are mounted as shown on the applicable drawing specified in 3.3.2.1.
(b) Parts, modules and drawers are marked with identifying symbols in accordance with the applicable drawing specified in 3.3.2.1.
(c) Terminals of parts and modules and terminal boards are marked with identifying letters, numbers, or both in accordance with the applicable drawing specified in 3.3.2.1.
(d) Wire used for interconnecting parts, modules and drawers are of the type shown on the applicable drawing specified in 3.3.2.1.
(e) There are no splices in any of the wires as specified in 3.3.1.
(f) Wire groups from hinged doors and panels are formed and clamped in a manner that sharp bends do not occur with the panel or door in the open or closed position as specified in 3.3.1.
(g) Sufficient slack in wiring is allowed so that the weight of the harness is not supported by the terminal connections; so that at least two replacements can be made if lugs are clipped off at the wire end as specified in 3.3.1.
(h) Cable entrance provisions and connections are as shown on the applicable drawing specified in 3.3.2.1.
(i) Sufficient space is provided for these connections and sufficient cable support within the drawers and cabinets, and connections made at the appropriate terminals as specified in 3.3.1.
(j) Wherever wires run through holes in metal partitions or chassis, grommets are provided for mechanical protection as specified in 3.3.1.
(k) Wires are not bent around sharp corners which may injure the insulation as specified in 3.3.1.
(l) Wires are connected by either bolted or soldered connections.

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are a part of the manufacturer's standard commercial products shall be

(m) Bolted connections are provided with locking devices as specified in 3.3.1.
(n) Both ends of all wires are marked with designations as shown on the applicable drawing specified in 3.3.2.1.
(o) The equipment pull-out drawers have stops to lock in pull-out position as shown on applicable drawings specified in 3.3.2.1.
(p) Non-identical drawers cannot be interchanged in the equipment cabinet as shown on applicable drawings specified in 3.3.2.1.
(q) Identification and instruction plates are furnished as shown on the applicable drawing specified in 3.3.2.1.

4.7 Tests.

4.7.1 Insulation resistance tests. Insulation resistance tests shall be conducted in accordance with method 302 of MIL-STD-202. Test conditions shall be as follows:

(a) Test potential: Test condition B (high internal impedance).
(b) Points of measurement: Between each electrically isolated circuit and all other circuits and ground (see 4.7.1.1 and figure 8).
(c) Electrification time: 60 seconds minimum.

4.7.1.1 Electrically isolated circuits shall be determined by application of the following criteria:

(a) Circuits, whose only connection to each other is by electromagnetic coupling through a magnetic core which is shared in common by the circuits, shall be considered to be electrically isolated from each other.
(b) Circuits, whose only connection to each other is through a capacitor, shall not be considered to be electrically isolated from each other. When the purpose is to test circuits internal to the equipment, and only then, such circuits shall be temporarily interconnected with a jumper wire or test lead.

4.7.2 Dielectric withstanding voltage tests. Dielectric withstanding voltage tests shall be conducted in accordance with method 301 of MIL-STD-202. Test conditions shall be as follows:

(a) Magnitude of test voltage.

(1) For circuits rated 60 V or less, the test voltage shall be 900 V.
(2) For circuits rated more than 60 V but not greater than 600 V, the root mean square (rms) test voltage shall be twice rated circuit voltage plus 1000 V.
(3) For circuits rated above 600 V, the rms test voltage shall be two and one-quarter times rated circuit voltage, plus 2000 V.

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(4) For circuits containing parts that are applied within specified ratings and are in accordance with part specifications that specify a lower dielectric test voltage than as specified in (1), (2) or (3) above, the dielectric test voltage for the circuit shall correspond to that specified for the part having the lowest specified dielectric test voltage but in no case less than:

a. 900 V for circuits connected to the power supply terminals.
b. 500 V for circuits electrically isolated from the power supply terminals.

(b) Nature of test voltage: The test voltage shall approximate a true sine wave of a frequency not less than the rated frequency of the circuit under test.

(c) Points of application: The test voltage shall be successively applied between each electrically isolated circuit and all other circuits and ground (see 4.7.1.1 and figure 8).

(d) Duration of test: The duration of the test shall be 60 seconds minimum. The test voltage shall be removed by adjustment of its value to zero, not by sudden interruption. An alternate test of 1 second minimum, using a touch probe, may be substituted if a test voltage 20 percent higher than that specified in (a) is used and a determination has been made that no surges which could damage parts are produced by sudden interruption of the test voltage.

(e) Definition of failure: Any evidence of arcing, flashover or punctured insulation shall be failure of the test. Corona (visible, audible or odorous) shall not be considered as a failure.

4.7.3 Self-protection from input voltage variation and transient. Test shall be performed as follows:

(a) The equipment shall be operated with nominal voltage at 20 and 40 percent of rated-load, 0.8 p.f. respectively. The input voltage to the equipment shall be raised and lowered 10 percent beyond the specified operation range for 1 second and 10 ms, respectively. The input voltage, output voltage and output current shall be recorded with an oscillograph.

(b) The equipment shall be operated at nominal input voltage to supply rated-load, interrupt the input power for a period of 30 seconds, and then restore to the original condition. The test shall be repeated by interrupting the input power for a period of 100 and 10 ms, respectively. The equipment input voltage, output voltage and output current shall be recorded with an oscillograph. The time to restore equipment output voltage, the magnitude of changes in output voltage and frequency, and the time required for the output voltage to return to the regulation band shall be determined. The oscillographic minimum deflection settings shall be as follows:

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(a) Output voltage = 2.5 inch deflection.
(b) Output current = 1 inch deflection at rated-load.
(c) Input voltage = 2.5 inch deflection.
(d) Chart speed = 10 inches per second for all tests except 1 inch per second for the 30 second interruption.

4.7.4 Self-protection from input spikes. The input to the equipment shall be nominal having a source impedance of 0.4 ohm inductive reactance (at 60 Hz) and 0.07 ohm resistance. The equipment shall be disconnected from the input lines and applicable transient voltage, as specified in 3.2.1.2, and shall be applied between line-to-line and between line-to-ground, respectively. After reconnecting the equipment back to the input lines, the transient voltage shall be reapplied as previously established when the equipment is operated at no-load and rated-load, 0.8 p.f. The equipment output voltage shall be observed on a memory oscilloscope with a complete 400 Hz waveform (both the positive and the negative half) on the screen. Pictures of the voltage waveforms shall be taken. The output voltage spikes shall be determined.

4.7.5 Input current harmonics. The equipment shall be operated at nominal input voltage and frequency. The input current harmonics of each input line shall be determined at no-load, 50 percent load and rated-load. Harmonics less than 0.1 percent may be disregarded. The current harmonics shall be measured by shunts connected in each input line to the equipment. The fundamental and harmonics of the voltage drops across the shunts shall be determined by a harmonic analyzer. Oscilloscope pictures of the input current waveform of each line at 50 percent load and at rated-load shall be taken.

4.7.6 Starting current and starting time. The nominal voltage shall be connected to the equipment input and the equipment shall be started with no-load at the equipment output. The input starting current of all three phases and output voltage shall be measured with an oscillograph. The test with full-load connected to the equipment shall be repeated. The equipment startup shall be attempted by setting the input voltage 2 V above and below the specified range and the input frequency 2 Hz above and below the specified limits in 3.2.1.5.1. The starting time and starting current shall be determined. An attempt to energize the unit shall be made when the input voltage phase rotation is in reverse order (AC-CB-BA) to determine any damage to the unit. The oscillograph minimum deflection settings shall be as follows:

(a) Input current = 2.5 inch deflection at rated-load.
(b) Output voltage = 1 inch deflection
(c) Chart speed = 5 inches per second for the starting current and 1 inch per second for the starting time.

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4.7.6.1 The equipment shall also start with the input voltage and frequency set at the following settings to ensure that the equipment will start in accordance with type I power of DOD-STD-1399, section 300:

(a) 410 V at 60 Hz
(b) 390 V at 60 Hz
(c) 470 V at 60 Hz
(d) 490 V at 60 Hz
(e) 440 V at 55 Hz
(f) 440 V at 57 Hz
(g) 440 V at 63 Hz
(h) 440 V at 65 Hz

4.7.7 Voltage and frequency regulation. The equipment shall be connected to an input voltage which can be varied as shown in table XIII. After the equipment has been stabilized at rated-load, the output steady-state voltage and frequency regulation shall be determined for all conditions specified in table XIII.

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TABLE XIII. Conditions of input and output.

Test condition numbers	Input						Output (of rated voltage and frequency)							
	Voltage within the specified range			Frequency within the specified range			Rated load				P.f.			
	max	nom	min	max	nom	min	0	50 percent	100 percent	125 percent	0.6 lag	0.8 lag	1.0	0.9 leading
1	X	-	-	X	-	-	X	-	-	-	-	-	-	-
2	-	X	-	-	X	-	X	-	-	-	-	-	-	-
3	-	-	X	-	-	X	X	-	-	-	-	-	-	-
4	X	-	-	X	-	-	-	X	-	-	X	-	-	-
5	-	X	-	-	X	-	-	X	-	-	X	-	-	-
6	-	-	X	-	-	X	-	X	-	-	X	-	-	-
7	X	-	-	X	-	-	-	X	-	-	-	X	-	-
8	X	-	-	-	X	-	-	X	-	-	-	X	-	-
9	-	X	-	-	X	-	-	X	-	-	-	X	-	-
10	-	-	X	-	X	-	-	X	-	-	-	X	-	-
11	-	-	X	-	-	X	-	X	-	-	-	X	-	-
12	X	-	-	X	-	-	-	X	-	-	-	-	X	-
13	X	-	-	-	X	-	-	X	-	-	-	-	X	-
14	-	X	-	-	X	-	-	X	-	-	-	-	X	-
15	-	-	X	-	X	-	-	X	-	-	-	-	X	-
16	-	-	X	-	-	X	-	X	-	-	-	-	X	-
17	X	-	-	X	-	-	-	X	-	-	-	-	-	X
18	-	X	-	-	X	-	-	X	-	-	-	-	-	X
19	-	-	X	-	-	X	-	X	-	-	-	-	-	X
20	X	-	-	X	-	-	-	-	X	-	X	-	-	-
21	-	X	-	-	X	-	-	-	X	-	X	-	-	-
22	-	-	X	-	-	X	-	-	X	-	X	-	-	-
23	X	-	-	X	-	-	-	-	X	-	-	X	-	-
24	X	-	-	-	X	-	-	-	X	-	-	X	-	-
25	-	X	-	-	X	-	-	-	X	-	-	X	-	-
26	-	-	X	-	X	-	-	-	X	-	-	X	-	-
27	-	-	X	-	-	X	-	-	X	-	-	X	-	-
28	X	-	-	X	-	-	-	-	X	-	-	-	X	-
29	X	-	-	-	X	-	-	-	X	-	-	-	X	-
30	-	X	-	-	X	-	-	-	X	-	-	-	X	-
31	-	-	X	-	X	-	-	-	X	-	-	-	X	-
32	-	-	X	-	-	X	-	-	X	-	-	-	X	-
33	X	-	-	X	-	-	-	-	X	-	-	-	X	-
34	-	X	-	-	X	-	-	-	-	X	X	-	-	-
35	-	X	-	-	X	-	-	-	-	X	-	X	-	-
36	-	X	-	-	X	-	-	-	-	X	-	-	X	-

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4.7.8 Output voltage remote sensing. The unit shall be connected to a rated-load with conductors which have a voltage drop of 7 percent of the rated output voltage at 125 percent of rated-load. With the unit operated at rated-load, 0.8 p.f., the voltage at the remote sensing point shall be adjusted to the rated voltage by means of the voltage adjustment; no further adjustment shall be allowed for the rest of this test. The voltage at the unit output terminals and at the remote sensing point shall be measured for test conditions 2, 5, 7, 9, 11, 14, 18, 21, 23, 25, 27, 30 and 35 of table XIII. The loss of remote sensing signal shall be simulated for each test condition. Voltage regulation, voltage modulation and voltage waveform at the remote point shall be determined. The voltage adjustment test shall be conducted as specified in 4.7.9 to determine the required voltage range at the remote sensing point.

4.7.9 Voltage adjustment. After the equipment has been stabilized at rated-load, the output voltages at the equipment output terminals shall be adjusted over their full ranges using the voltage adjustment devices. The test shall be conducted for test conditions 2, 5, 7, 9, 11, 14, 18, 21, 23, 25, 27, 30 and 35 of table XIII. The voltage at the unit output test points shall also be measured with a voltmeter.

4.7.10 Transient voltage and frequency deviation and recovery. With the equipment initially operating with nominal input, a balanced 3-phase load of 0.8 p.f. shall be changed in one step for the following conditions:

(a) From no-load to half-load
(b) From half-load to rated-load
(c) From rated-load to half-load
(d) From three-fourths load to one-fourth load
(e) From no-load to rated load
(f) From rated-load to 5 percent load
(g) From rated-load to no-load

Oscilloscope pictures shall be taken of the load application and removal to deter- the voltage transient and recovery time. This test shall be conducted for input test conditions 5, 8, 10, 11, 13, 14 and 15, of table XIII. The frequency transient and recovery test shall be performed the same way except that load application and removal shall be from no-load to rated-load and from rated-load to no-load in one step, respectively.

4.7.11 Amplitude and frequency modulation. Amplitude frequency modulation of the equipment shall be determined at the equipment output terminals for the test conditions 1, 3, 5, 7, 9, 11, 12, 14, 16, 21, 23, 25, 27, 28, 30, 32 and 35 of table XIII. The voltage amplitude shall be measured with an oscilloscope. The oscilloscope settings shall be 10 V per division and 10 or 20 ms per division. Oscilloscope pictures shall be taken.

4.7.12 Output voltage waveform. The harmonic content and output voltage spikes shall be determined with the equipment operating under the test conditions 2, 5, 9, 14, 21, 25 and 30 of table XIII. Oscilloscope pictures of the waveform shall be taken.

4.7.13 Output phase displacement angle. The equipment output displacement angle shall be calculated for test conditions 2, 5, 9, 14, 18, 21, 25 and 30 of table XIII.

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4.7.14 Output voltage balance under unbalanced load. The equipment shall be operated with nominal input and the equipment output phases (three phases) loaded to produce 80 percent of the rated current at 0.6 p.f. A single phase 0.6 p.f. load shall be applied across the equipment output phase A-B to have line A carry an additional current of 7 percent of rated current. The output voltage of all phases shall be recorded at the equipment output terminals. The test shall also be conducted, using a single phase 0.6 p.f. load across equipment output phase B-C and the C-A. The equipment shall be operated at nominal input. A single phase, 0.6 p.f. load shall be applied across equipment output phases A-B, to have line A carry a current of 7 percent of the rated current. The output voltages of all phases shall be recorded. The test shall also be conducted, using a single phase 0.6 p.f. load across equipment output phases B-C and then C-A. The above tests shall be repeated for 0.8 p.f. load. The output voltage regulation shall be determined.

4.7.15 Protection for utilization equipment. The equipment shall be operated at nominal input at no-load. Voltage and frequency signals shall be injected into the equipment sensing circuits to simulate the conditions of equipment output overvoltage, undervoltage, overfrequency and underfrequency as specified in table VI. The output voltage and frequency and the time required for the equipment to shut down shall be recorded with an oscillograph. The oscillograph minimum deflection settings shall be as follows:

(a) Output voltage = 2 inch deflection.
(b) Output current = 1 inch deflection at rated-load.
(c) Chart speed = 10 inches per second.

4.7.16 Input voltage change effect. The equipment shall be operated at 440 V, 60 Hz input. The input voltage and frequency shall be increased and then decreased in one step to the amount specified in 3.2.2.12. The input and output voltages shall be recorded by oscillograph, to determine the output voltage transient and recovery time. The test shall be conducted at no-load and rated-load, 0.8 p.f. The oscillograph minimum deflection settings shall be as follows:

(a) Output voltage = 2.5 inch deflection.
(b) Output current = 1 inch at rated-load.
(c) Input voltage = 2.5 inch deflection.
(d) Chart speed = 10 inches per second.

4.7.17 Efficiency and no-load input current. Meters shall be inserted in the input and output for reading voltage, current and wattage. Readings shall be recorded for test conditions 2, 9, 10, 14, 15, 25, 26, 30 and 31 of table XIII. Efficiency shall be calculated according to the following formula:

$$\text{Percent efficiency (overall)} = \frac{\text{watts (output)} \times 100}{\text{watts (input)}}$$

This test shall be repeated for unity p.f. load. Heat dissipation to ambient shall be determined for water-cooled units. No-load losses shall also be determined.

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4.7.18 Rating. The equipment shall be operated at nominal input for 2 hours each at the following loads:

(a) Rated-load, 0.6 p.f. lagging.
(b) Rated-load, 0.8 p.f. lagging.
(c) Rated-load, unity p.f.
(d) Rated-load, 0.9 p.f. leading.

Input power, output power, output current, output frequency and output voltage shall be recorded at the beginning and at the end of each 2-hour load run.

4.7.19 Input current balance. The input voltage to the equipment shall be nominal with 3 percent unbalance among the three phases. The equipment shall be operated at 10 percent load, 50 percent load and rated-load at rated output voltage and frequency. The input current of each line to the equipment shall be measured with indicating meter having initial accuracy of 1 percent or better. This test shall be repeated with the input 3-phase voltages set at the nominal value and balanced within 4 V.

4.7.20 Overload. The equipment shall be operated at nominal input and rated output, and then overloaded to 125 percent of rated-load at rated output voltage 0.8 p.f. for more than 1 minute. The output voltages, output currents, output amplitude modulation and time required for the equipment to shutdown shall be determined by an oscillograph. The oscillograph minimum deflection settings shall be as follows:

(a) Output voltage = 2 inch deflection.
(b) Output current = 1 inch deflection.
(c) Chart speed = 0.5 inch per second.

4.7.21 Short circuit and fault current. A 3-phase and single phase short circuit fault shall be applied to the equipment output terminals. This test shall be conducted with input voltage of maximum, nominal and minimum in the specified range. The fault current and the time required to shut down the equipment shall be recorded with an oscillograph. The above test shall be repeated with a 3-phase fault current of 130 percent of the rated-load current. The oscillograph minimum deflection settings shall be as follows:

(a) Output voltage = 1 inch deflection.
(b) Output current = 2 inch deflection.
(c) Chart speed = 0.5 inch per second.

The short circuit test shall be repeated with the short circuit taken off the unit 25 $\pm$ 5 seconds after application.

4.7.22 Parallel operation. Three units (two units for quality conformance tests) of the same or different kW ratings shall be operated in parallel. Two (one) units shall be supplied from one input power source and the other unit from a different input power source. After the units have been automatically synchronized to operate in parallel, no further adjustment shall be made for the remainder of this test. The total load of the three (two) units shall be varied from 10 to 90 percent of the combined kW and kVA rating of the three (two) units at 0.8 p.f. in 25 percent load steps and back to 10 percent load in the same manner. The units shall operate at each load step for a minimum of 60 minutes.

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The difference in kW and kVA supplied by each unit shall be recorded and computed. For each load step, the input voltage and frequency of one source shall be maintained constant and the other source shall be varied as follows:

Constant input	Variable input	Minimum operation time
440 V, 60 Hz	440 V, 60 Hz	10 minutes
440 V, 60 Hz	462 V, 60 Hz	10 minutes
440 V, 60 Hz	418 V, 60 Hz	10 minutes
440 V, 60 Hz	418 V, 57 Hz	10 minutes
440 V, 60 Hz	462 V, 63 Hz	10 minutes

Simulated internal fault shall be applied to one of the paralleled units to demonstrate selective tripping. The unit output voltage shall be recorded by an oscillograph or picture taken from an oscilloscope. The parallel operation test shall be conducted for both the common bus arrangement and the split bus arrangement. The input voltage, input current, input power, input frequency, output voltage at each bus, output frequency, output power and output current shall be recorded for each load condition. The voltage regulation and load sharing shall be determined. When the paralleled units are operated with nominal input voltage at no-load, at about 50 percent load and at 90 percent load, the following tests shall also be conducted to determine system stability and selective tripping:

(a) Apply simulated internal faults to the drawers of the units (at 50 percent load only).
(b) Manually turn off each one of units in turn. Start up the turned-off unit and put it back in parallel with the other two units.
(c) Interrupt the input power to each unit in turn. Reapply the input power to the turned-off unit and put it back in parallel with the other two units.
(d) Transfer the control from "local" to "remote" and vice versa.
(e) Apply a short circuit to the bus (for common bus) or one of the buses (for split bus) for a period of 1 to 5 seconds and then remove the short circuit.

In the above tests, a, b, c and d, an oscillograph shall be used to record the bus voltage, load current and output current from each unit.

4.7.23 Output voltage balance under balanced load. After the equipment has been stabilized at rated-load, the output voltage of each phase at the equipment output terminals shall be determined for the test conditions 2, 5, 9, 14, 18, 21, 25 and 30 of table XIII.

4.7.24 Shock. High-impact shock tests (electrical operation and mechanical damage) shall be conducted in accordance with MIL-S-901 and as specified in 4.7.24.1.

4.7.24.1 Test features. Test features shall be as follows:

(a) Type A (for principal units) test shall be performed. The equipment shall be energized and operated at 10 percent of rated-load during shock test.

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(b) Definition of failure to perform principal functions shall be as follows:

(1) Breakage of any parts, including mounting bolts: Minor chipping of parts such as plastic knobs and cases and minor distortion of parts will be permitted where such chipping or distortion cannot in any manner impair operation of the equipment as specified.
(2) Appreciable distortion of any parts, including enclosure and framework.
(3) A value of insulation resistance lower than as specified (see 3.3.2.2).
(4) Low dielectric strength: After shock tests, the dielectric test shall be conducted at a voltage equal to 65 percent of the voltage specified in 4.7.2. Failure to pass this test shall be cause for rejection.
(5) Failure to pass visual examination: The equipment shall be examined after removing all removable panels, doors and drawers to ascertain any mechanical damage. If necessary, partial disassembly shall be performed to aid in determining any possible damage.
(6) Failure to continually carry rated-load during shock test and to perform electrical test following shock. No adjustment of replacement of damaged parts shall be permitted after the shock test and prior to the tests specified in 4.7.24.1(d).

(c) Mounting: Equipment shall be mounted on the shock machines in a manner simulating shipboard installations in accordance with MIL-S-901 for deck mounted equipment.
(d) Immediately following the shock test, the equipment shall be tested for output voltage and frequency regulation (see 4.7.7) and transient voltage and frequency deviation and recovery (see 4.7.10) with only nominal input.

4.7.25 Vibration. Vibration tests shall be conducted in accordance with type I of MIL-STD-167-1. The load to the equipment for the vibration test shall be as specified for the shock test (see 4.7.24).

4.7.25.1 Test feature. The failure determination as specified for shock tests in 4.7.24.1(b) shall apply also to vibration tests. The equipment selected for vibration test shall be the same one selected for high-impact shock tests.

4.7.26 Structureborne noise. The structureborne noise shall be determined in accordance with MIL-STD-740. For the noise tests, the equipment shall be operating at full-load. The contractor shall prepare test reports in accordance with the data ordering document (see 6.2.2).

4.7.27 Airborne noise. The airborne noise shall be determined in accordance with MIL-STD-740. Load conditions during tests and reports of tests shall be the same as specified for structureborne noise tests (see 4.7.26).

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4.7.28 Cooling system failure protection test. The equipment shall be operated at nominal voltage input and at rated-load. The cooling fan circuit shall be interrupted to simulate fan failure. The equipment input and output voltage shall be observed with an oscilloscope. For water-cooled equipment, the cooling water shall also be shut off to simulate cooling water failure.

4.7.29 Electromagnetic interference (EMI) emission and susceptibility. An electromagnetic interference emission and susceptibility test shall be conducted in accordance with MIL-STD-462 for CE01, CE03 and RE02. Tests shall be conducted on each input and output conductor and remote control circuit conductor of the equipment. Data for both the broadband and narrowband of the conducted emission levels shall be taken and incorporated in the EMI test report in accordance with the data ordering documents (see 6.2.2).

4.7.30 Temperature rise and high temperature operation. Temperature rise tests shall be conducted at an ambient temperature of 50°C. The test methods employed and the precautions to be observed shall be as specified in 4.7.30.1 through 4.7.30.5.

4.7.30.1 Assembly and mounting of equipment. Temperature rise tests shall be made only with the equipment completely assembled and mounted in the manner for which it is designed. Barriers shall be placed adjacent to the enclosure to simulate shipboard space restrictions. The minimum temperature measurement points shall be as follows:

(a) Power semiconductor studs.
(b) Capacitors.
(c) Power transformer and reactor coil windings.
(d) Ambient temperature.
(e) Cooling air inlet and outlet.

4.7.30.2 Method of temperature measurements. The methods of ambient temperature measurement of parts or subassemblies shall be in accordance with MIL-E-917 except that thermometer shall not be used.

4.7.30.3 Method of loading. The equipment shall be operated at test conditions 29, 30 and 31 of table XIII. The temperatures shall reach the steady final value as specified in 4.7.30.4 for each input voltage and for each load condition. Temperature rises shall be in accordance with temperature requirements of MIL-E-917.

4.7.30.4 Duration of test. The consecutive temperature measurement of each part or module shall be recorded not more than 5 minutes apart. The temperature rise test shall be continued until the temperature rise of parts or modules that can be observed or recorded during test, has attained a steady final value. Steady final value is temperature which will remain at a 2°C band for a period of 2 hours, while the ambient temperature is maintained at 50°C. The incoming cooling water (for the water-cooled unit) shall be maintained in the range of 29 to 35°C.

4.7.30.5 High temperature operation. After the equipment has attained a steady final value as specified in 4.7.30.4, the following tests shall be conducted at the ambient temperature of 50°C (except that load conditions shall be as specified in 4.7.30.3):

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(a) Voltage and frequency regulations test as specified in 4.7.7.
(b) Voltage adjustment test as specified in 4.7.9.
(c) Transient voltage and frequency deviation and recovery test as specified in 4.7.10.
(d) Amplitude and frequency modulation test as specified in 4.7.11.
(e) Output voltage waveform test as specified in 4.7.12.
(f) Short circuit protection test as specified in 4.7.21.
(g) Input voltage step change test as specified in 4.7.16.

4.7.31 Grounding. For the grounding test, equipment shall operate at nominal input voltage and at no-load and rated output load. Each input and output terminal shall be grounded in turn to ground as follows:

(a) Connect each terminal to ground through a 100K ohm resistance.
(b) Momentarily ground each terminal in turn five times by shorting one of the 100K ohm resistors in (a). Remove that short and then short, in consecutive order, each of the other 100K ohm resistors in (a). Also ground each terminal for a period of 10 seconds and then remove the short.

The oscillograph shall be used to determine any voltage changes in both the equipment input and output. The oscillograph minimum deflection settings shall be as follows:

(a) Input voltages = 2 inch deflection.
(b) Output voltages = 2.5 inch deflection.
(c) Chart speed = 1 inch per second.

4.7.32 Humidity. The equipment shall be tested as specified in 4.7.32.1, 4.7.32.2, and 4.7.32.3.

4.7.32.1 Procedure. Transformer and reactor windings shall be disconnected as necessary from other components to permit accurate measurements of insulation resistance and dissipation factor of the winding and ground insulation. Initial measurements of insulation resistance and dissipation factor shall be taken as well as the temperature and relative humidity of the surrounding air. The equipment shall then be conditioned for a period to arrive at a dry condition. Conditioning shall be obtained by placing the wound components in a heat chamber of drying oven for a period of 6 hours with temperature maintained at 105°C. At the end of this period the wound components shall be removed and after cooling (within 8 hours) the insulation resistance and dissipation factor shall again be measured. The equipment shall then be immediately placed in the humidity chamber for a period of 1 week (168 hours). The humidity cycle shall be in accordance with the curve as specified on figure 9. While under humidification, daily measurements of ambient temperature, insulation resistance and dissipation factor shall be made 1 hour after the humidity chamber temperature has been raised to 50°C and 1 hour before the humidity chamber temperature has been lowered to 30°C. The insulation resistance and dissipation factor shall meet the values specified on figures 10 and 11. During the 7 days of humidification cycle, the drawers shall be opened to permit free access of the humidified air to all parts and surfaces. At the end of the 160-hour humidity period, the equipment shall be carefully examined for any signs of material damage and corrosion. The extent of any observed damage or corrosion shall be noted and cause shall be determined. After examination, any disconnections or disassembly shall be restored to the

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original condition, and the equipment output voltage and frequency regulation and voltage adjustment tests as specified in 4.7.7 and 4.7.9 shall be rerun. Also following operation, insulation resistance shall be measured and recorded between isolated circuits and between circuits, and grounded as specified in 4.7.1. These tests shall be conducted within 8 hours after the 160-hour humidity period, at 50 percent humidity and 25°C.

4.7.32.2 Recommended test apparatus. Recommended test apparatus shall be as follows:

(a) The insulation resistance measuring equipment shall be similar and equivalent in accuracy to the General Radio Megohm Bridge, type 544-B, 500 V d.c. with a range from 0.1 to 1,000,000 megohms.
(b) The dissipation factor measuring equipment shall be similar and equivalent in accuracy to the General Radio Capacitance Test Bridge, type 1611-A, 60 Hz with a range of dissipation factor from 0 to 50 percent.

4.7.32.3 Methods of measurement. Methods of measurement shall be as follows:

(a) The insulation resistance measurement shall be made using a potential of 500 Vd.c. applied for a 1-minute period. The insulation resistance shall be measured on all transformers and reactors. Measurements shall be made and recorded between each winding and ground and between insulated windings.
(b) The dissipation factor shall be measured with the capacitance test bridge with values read directly in percent from the bridge. Dissipation factor shall be measured on all transformers and reactors. Measurements shall be made and recorded between each winding and ground, and between insulated windings.

Due to the need for exposing copper for the connections to the windings, a leakage path is produced by humidification over the various insulation surfaces between the bare copper and other parts of the equipment that are otherwise insulated from the copper circuit. Therefore, leads shall be held separate as much as possible and all exposed terminals covered with a nonwetting grease which shall be wiped off with a clean dry rag upon removal from humidification. Insulation resistance and dissipation factor measurements during the humidification period shall be made with equipment in the humidity chamber.

4.7.33 Remote startup and shutdown. The equipment shall be operated at nominal input voltage. The control of the equipment shall be from a remote control panel which is connected to the equipment with a cable having an equivalent resistance of a 500 feet type MSCU cable in accordance with MIL-C-915/32. Equipment operation from the remote control panel and the transfer of control from the "local" to the "remote" control and vice versa shall be demonstrated.

4.7.34 Motor starting test. The equipment shall be operated at nominal voltage input. A 0.3 lagging p.f. static load shall be connected to the equipment through a circuit breaker or contactor to simulate a 400 Hz motor starting condition. The static load shall draw 125 percent of the equipment rated current. The circuit breaker or contactor shall be closed for a period of 30 seconds and the equipment output voltage and current shall be recorded with an oscillograph. The oscillograph minimum deflection settings shall be as follows:

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(a) Output voltage = 3 inch deflection.
(b) Output current = 1 inch deflection at rated-load.
(c) Chart speed = 1 inch per second.

4.7.35 Load transfer to another source. Equipment shall be operated at nominal input with rated-load. The load shall be transferred by a "make-before-break" switch to a motor generator set of not less than the rating of the equipment. The load transfer shall be accomplished with the output voltage of the equipment and the motor generator set in phase. Voltage waveform pictures shall be taken by oscilloscope or oscillograph.

4.7.36 Reliability test. The reliability test shall be performed in accordance with test plan IV A of MIL-STD-781, except as otherwise specified herein.

4.7.36.1 Reliability test conditions. The following reliability test conditions shall apply:

(a) Temperature - 45 to 50°C.
(b) Temperature cycling - None.
(c) Vibration - None.
(d) Output load cycling - During each 8-hour day shift, the unit shall be cycled three times. Each load cycle shall consist of the following:

(1) Rated-load - 1.5 hours.
(2) No-load - 20 minutes.
(3) Shutdown - 10 minutes.

During the remainder of each 24-hour day, except for test specified below, the unit shall be operated at rated-load.

(e) Output voltage - Rated output voltage.
(f) Output frequency - Rated output frequency.
(g) Input voltage - For load cycle during 8-hour day shift as specified in (d) above, the input voltage shall be as follows:

(1) The first load cycle - 420 $\pm$ 1 V.
(2) The second load cycle - 440 $\pm$ 1 V.
(3) The third load cycle - 460 $\pm$ 1 V.

For the remainder of each 24-hour day, the input voltage shall be 440 $\pm$ 22 V.

(h) Input frequency - 60 Hz.
(i) Incoming cooling water temperature - 29 to 35°C (for water-cooled unit).
(j) Additional test - After the third load cycle of each 8-hour day shift, the output voltage transient shall be conducted as specified in 4.7.10(a) and 4.7.10(b) with input voltage frequency at 440 V, 60 Hz and output voltage and frequency at 450 V, 400 Hz. Output voltage regulation and modulation shall be conducted as specified in 4.7.7 and 4.7.11 with input voltage and frequency at 440 V, 60 Hz and output voltage and frequency at 450 V, 400 Hz. The load conditions for output voltage regulation and

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modulation tests shall be a no-load, 50 percent load and rated-load, 0.8 p.f. lagging. The input current, input voltage, input frequency, output current, output frequency, output voltage and output power shall be recorded at rated-load and at no-load for each load cycle during each 8-hour day shift. For the remainder of each 24-hour day, these data shall be recorded by pen and ink type or equivalent recorders.

4.7.36.2 Failure. The definition of failure shall be in accordance with 3.5.1.

4.7.37 Low temperature operation. Before this test, the equipment shall be conditioned to ensure that each part is at a temperature of 0 to 5°C. A nominal input shall be connected to the equipment. Equipment shall be started at an ambient temperature of 0°C and shall operate at no-load and rated-load 0.8 p.f. for a period of 1 hour at each load condition.

4.7.38 Pulsing load and nonlinear load. The equipment shall be operated with the nominal input voltage. The output shall be connected to a rated-load which consists of 20 percent of 0.8 p.f. load and 70 percent of resistive load. The resistive load shall be connected through a load simulator which will be provided by the Navy. The load simulator shall be set to have steady load equal to 45 percent of rated-load and pulsing load equal to 25 percent of rated-load. The frequency of the simulator shall be manually adjusted from 1 to 50 Hz and the equipment output voltage amplitude modulation shall be monitored at the frequency which produces the highest amplitude modulation, data shall be taken to determine the output voltage regulation and amplitude modulation. The amplitude modulation shall be measured with an oscilloscope, having settings of 10 V per division and 10 or 20 ms per division. Oscilloscope pictures shall be taken. The pulsing load test shall be repeated with the load simulator set to have steady-state load equal to 10 percent of rated-load and pulsing load equal to 25 percent of rated-load. The nonlinear test shall be conducted at the nominal input voltage and with the output connected to a resistive load equal to 75 percent of ratedload through a 3-phase full wave rectifier bridge. The nonlinear load test shall be repeated with the output voltage sensing at the input of the rectifier bridge. The cable used to connect the equipment to the rectifier bridge, shall have a voltage drop of 7 percent of rated voltage at 125 percent of rated current. The equipment output voltage regulation, amplitude modulation and voltage waveform (harmonics, deviation factor and voltage spikes) shall be determined. Oscilloscope pictures of the voltage amplitude modulation and voltage waveform shall be taken.

4.7.39 Transient output voltage and recovery for electronic load. With the equipment operating at nominal input, a balanced 3-phase load of 125 percent rated-load at 0.6 p.f. shall be changed to no-load in one step. Oscillograms or oscilloscope pictures shall be taken of the load application and removal to determine the output voltage transient and recovery time.

4.7.40 Input power factor. The equipment shall operate at nominal input voltage, and rated output voltage and frequency. The input power in kW and kVA shall be recorded and the input power factor shall be determined for the no-load, 50 percent load, 0.8 p.f., and rated-load, 0.8 p.f. conditions.

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4.7.41 Endurance run. A continuous 168-hour endurance run shall be conducted on the equipment. The test conditions shall be as follows:

(a) Ambient temperature - For one of the three load cycles during each 8-hour day shift, the ambient temperature shall be in the range of 20 to 25°C. For the other two load cycles during each 8-hour day shift and the remainder of each 24-hour day, the ambient temperature shall be 45 to 50°C.
(b) Incoming cooling water temperature - 29 to 35°C (for water-cooled equipment only).
(c) Output load cycling - During each 8-hour day shift, the equipment shall be cycled 3 times. Each load cycle shall consist of the following:

(1) Rated-load - 1.5 hours.
(2) No-load - 20 minutes.
(3) Shutdown - 10 minutes.

During the remainder of each 24-hour day, the equipment shall be operated at rated-load.

(d) Output voltage - rated output voltage.
(e) Output frequency - rated output frequency.
(f) Input voltage - 440 ± 22 V.
(g) Input frequency - 60 Hz.
(h) Data - The input current, input voltage, input frequency, output current, output frequency, output voltage, and output power shall be recorded for each load cycle during each 8-hour day shift. For the remainder of each 24-hour day, these data shall be recorded by pen and ink type, or equivalent recorders.
(i) Additional tests - After completion of the 168-hour run, the output voltage transient shall be conducted as specified in 4.7.10(a) and 4.7.10(b), the output voltage regulation and amplitude modulation shall be conducted as specified in 4.7.7 and 4.7.11. For these tests, the input voltage, input frequency, output voltage and output frequency shall be the same as the endurance run. The load conditions for output voltage regulation and amplitude modulation tests shall be at no-load, 50 percent load and rated-load, 0.8 p.f. lagging.

4.7.41.1 Failure. The definition of failure shall be in accordance with 3.5.1. A failure is not considered to have occurred if malfunction can be attributed to external causes such as variation of input power parameters beyond specified limits, input power interruption, operator's error, or ambient temperature over 50°C.

4.8 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.8).

5.1 Preservation-packaging, packing and marking. The equipment shall be preserved-packaged in accordance with applicable drawings (see 3.3.2.1).

5.1.1 Special marking. In addition to the marking requirements of 5.1, equipment tested and certified (see 3.4.3, 4.7.26, and 4.7.27) shall have shipping containers marked in accordance with MIL-STD-740.

5.2 Tools. Tools shall be preserved-packaged, packed and marked in accordance with MIL-E-17555 or PPP-P-40, as applicable with the level specified for the intended use and destination as follows. Installation spares shall be as specified (see 6.2.1):

	Levels	
Destination	Preservation-packaging	Packing
On board (accompanying equipment)	A	C
Stock	A	B
Immediate use	C	C
INCO kits	A	C

6. NOTES

6.1 Intended use. Solid state frequency changers covered by this specification are intended to convert Navy shipboard 60 Hz electrical power into 400 Hz electrical power for surface ships, except for high performance ships, such as PHM, PCH, and so forth.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type (see 1.2).
(c) Whether first article inspection is required (see 3.1 and 4.4).
(d) Power rating (see 3.2.3).
(e) Mating electrical connectors (see 3.3.2.1).
(f) Installation spares (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirement List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR

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7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.1	Calibration system description	DI-R-7064	-----
4.4.1	First article inspection report	DI-T-4902	-----
4.5.1	Procedures, test	UDI-T-23732	-----
4.7.26 and 4.7.27	Summary, shop noise test results	UDI-T-23760	-----
4.7.29	Electromagnetic interference test report	DI-R-7062	-----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Equipment. Equipment is a solid state frequency changer having the power rating of 63 kW, 160 kW or 250 kW for water-cooled units and 72 kW, 180 kW or 285 kW for air-cooled units.

6.4.2 Unit. Unit is each power rating of the solid state frequency changers; for example, a 250 kW frequency changer is one unit and a 160 kW frequency changer is another unit.

6.4.3 Cabinet. Cabinet is a number of drawers physically joined together to perform a specific function.

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6.4.4 Drawer. Drawer is two or more parts or removable modules in one drawer. A drawer is a portion of a cabinet.

6.4.5 Module. Module is two or more parts to form an entity as part of a drawer. Module can be replaceable as a whole, but having a part or parts which are individually replaceable.

6.4.6 Parts. Parts is one piece or two or more pieces joined together which are not normally subject to disassembly, without destruction of intended use. The parts are the lowest level of breakdown.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Deviations. Deviations from the specified requirements and drawings are not permissible, without prior written authorization from the Naval Sea Systems Command. Request for the use of parts, other than those specified in the drawings, shall conform to procedure I of MIL-STD-965.

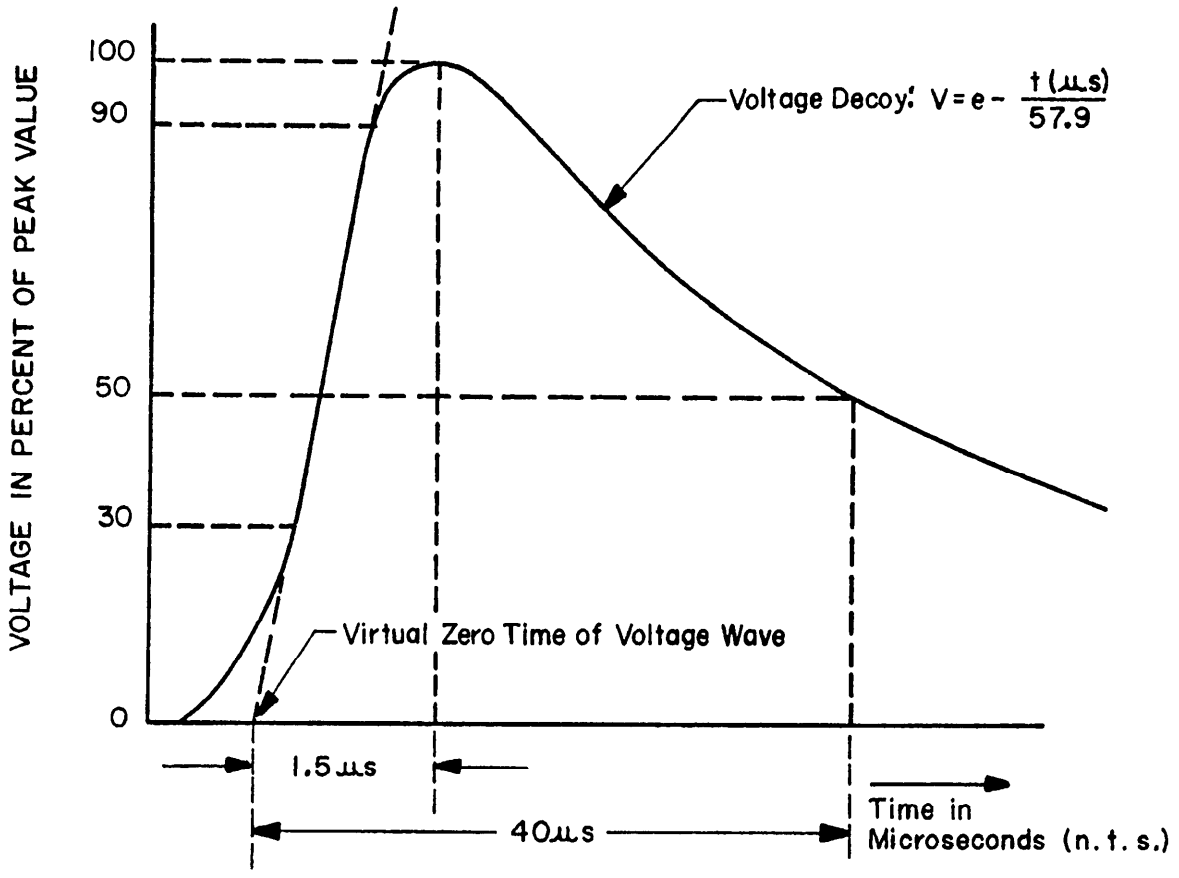
6.7 Frequency changer mounts. The frequency changer should be solidly mounted on surface ships with non-ASW capability and resiliently mounted for ASW capability.

6.8 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6130-N240)

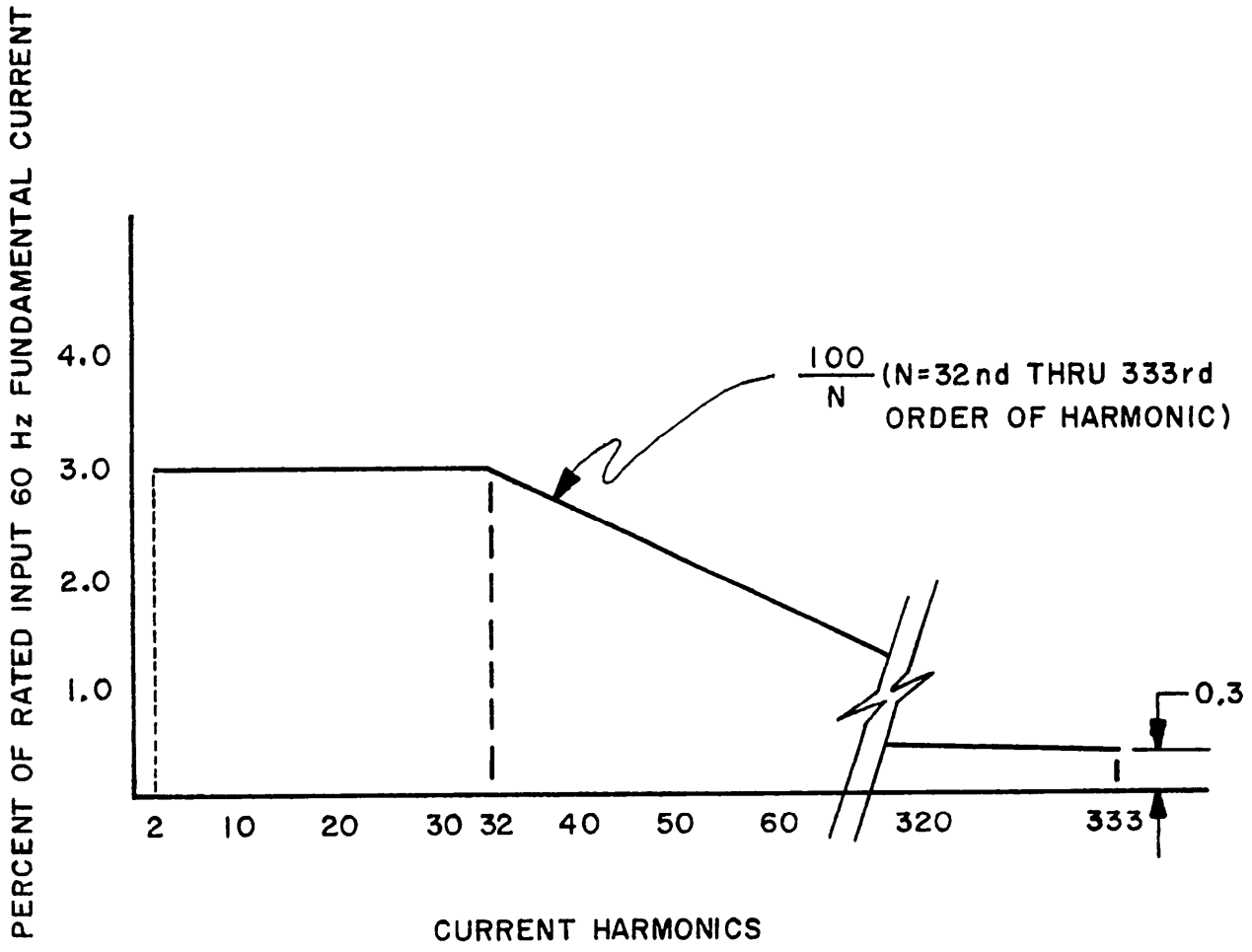
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SH 12424

FIGURE 1. Spike voltage (short time transient) wave shape.

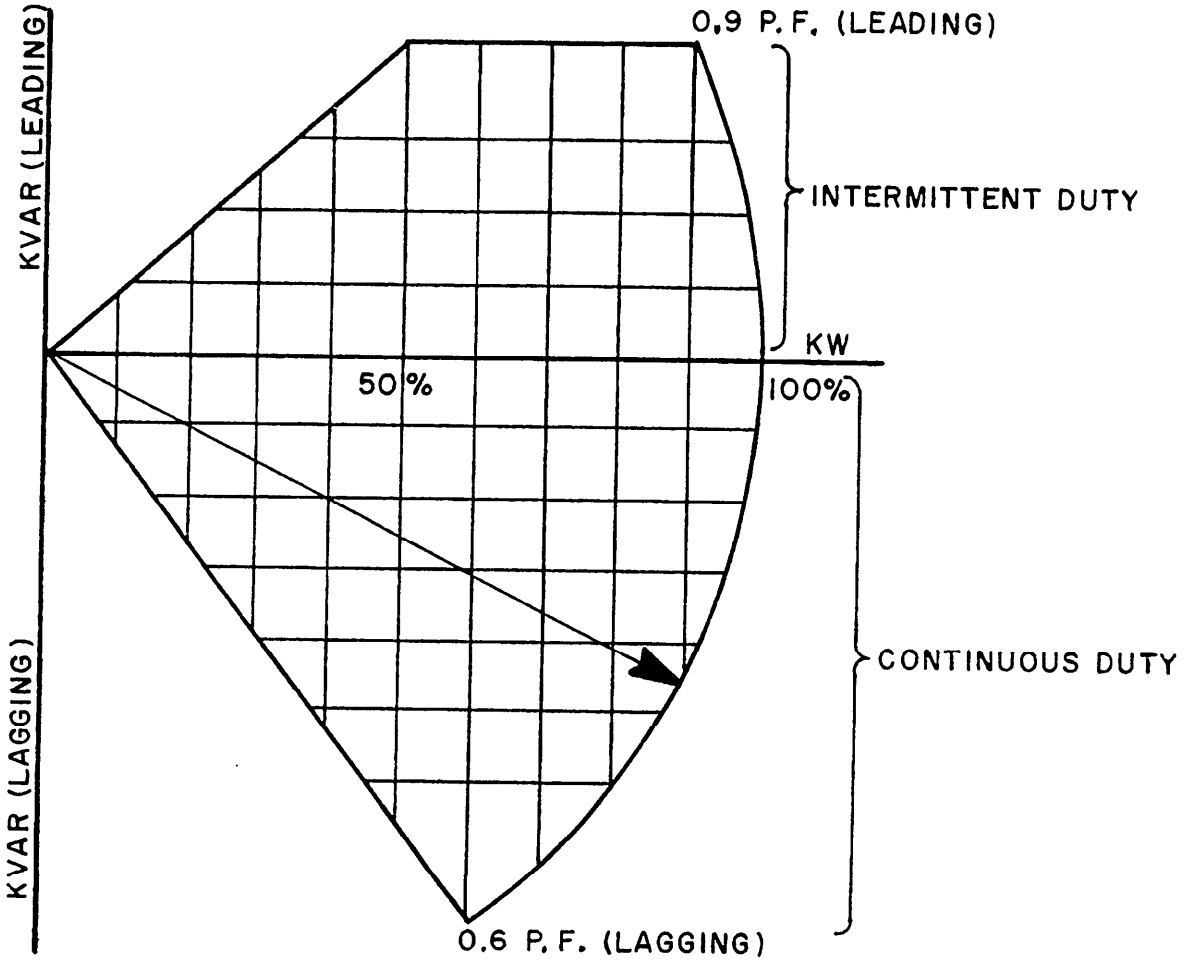
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SH 12425

FIGURE 2. Maximum allowable input current harmonics.

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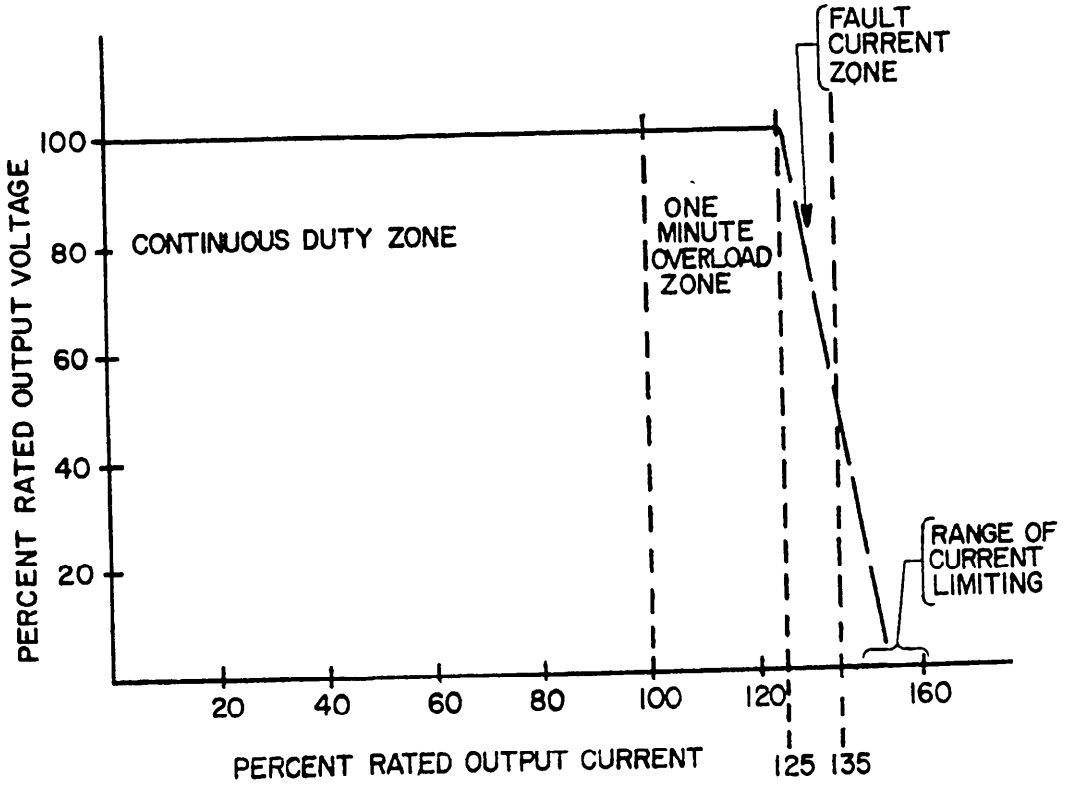


$$\text{RATED KVA} = \frac{\text{RATED KW}}{0.8}$$

SH 12426

FIGURE 3. Rating.

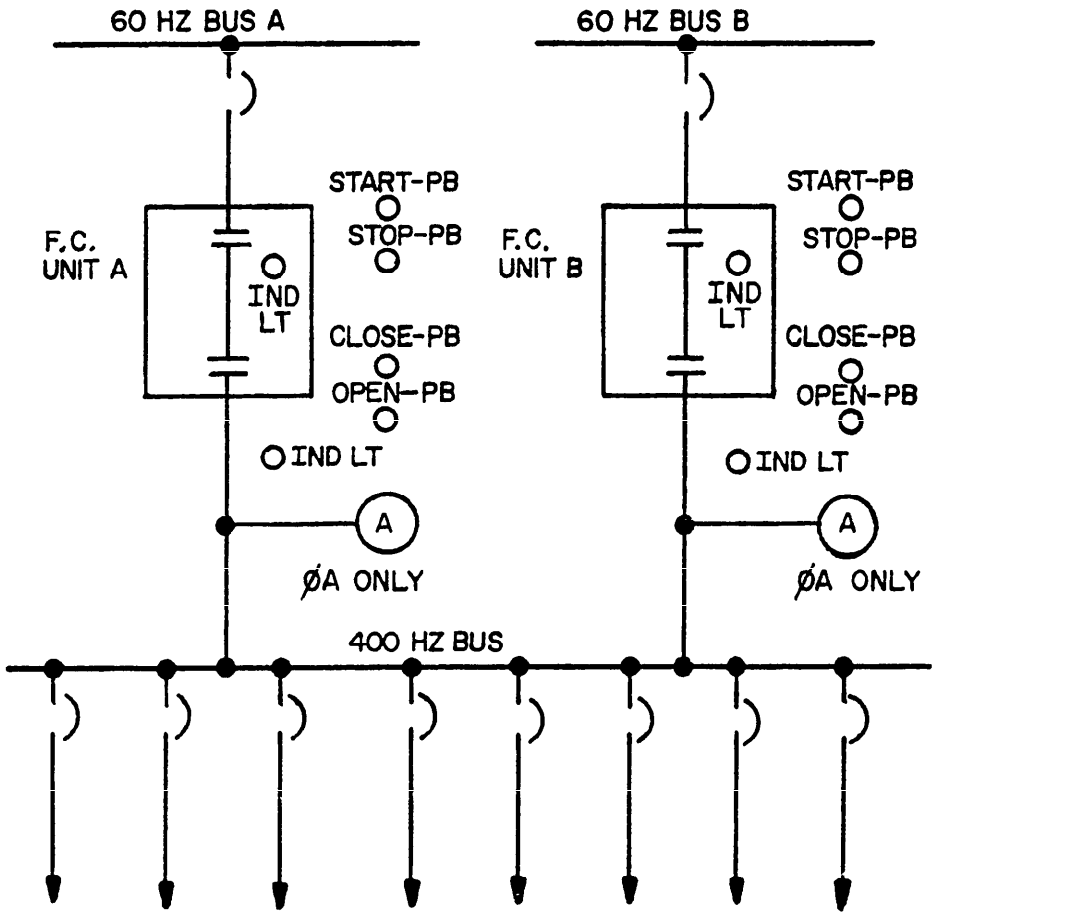
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SH 11286

FIGURE 4. Voltage versus current.

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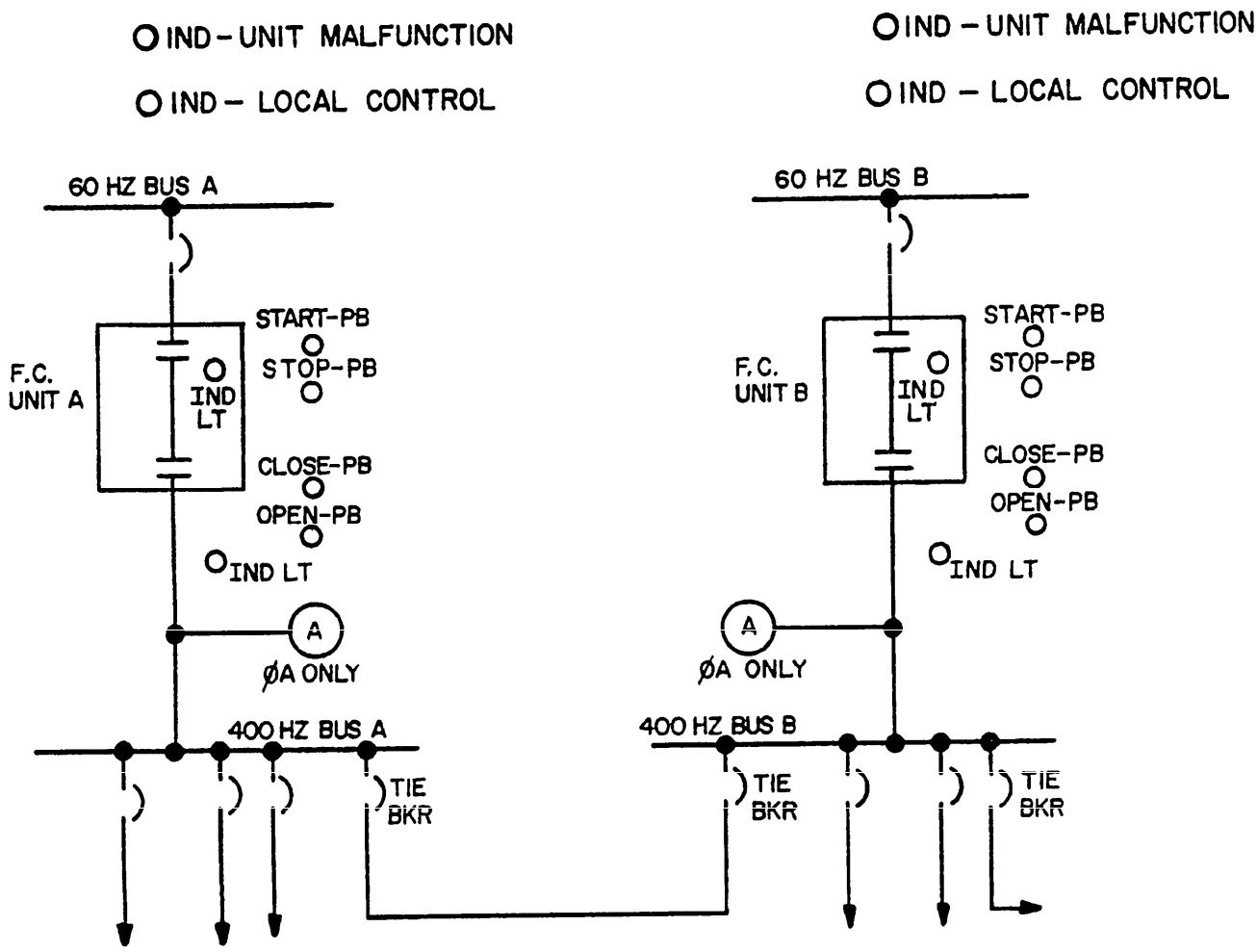


NOTE: Shown for two units only, but same arrangement applicable for up to six units.

SH 11287

FIGURE 5. Parallel operation - common bus.

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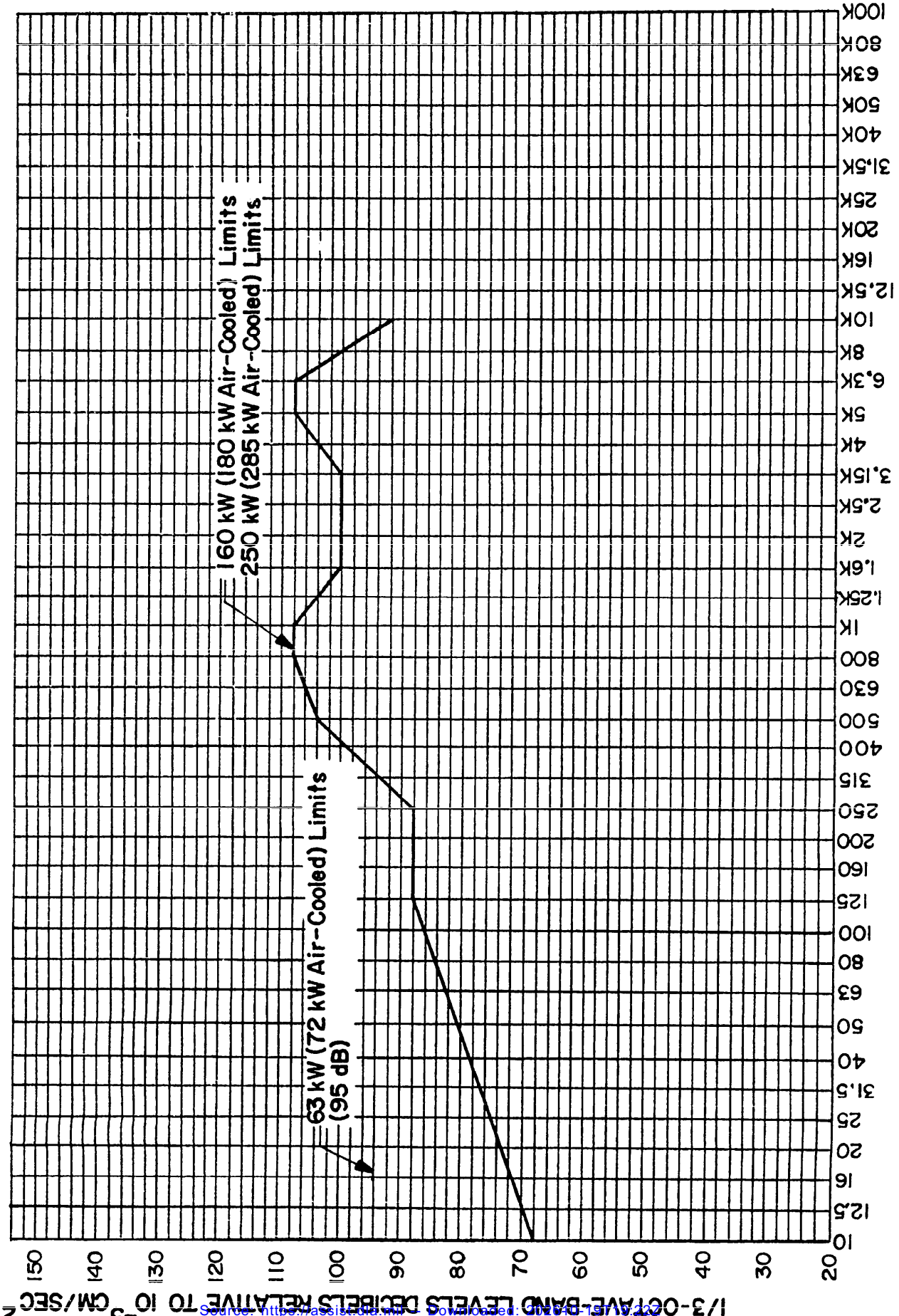


NOTE: Shown for two units only, but same arrangement applicable for up to six units.

SH 11288

FIGURE 6. Parallel operation - split bus.

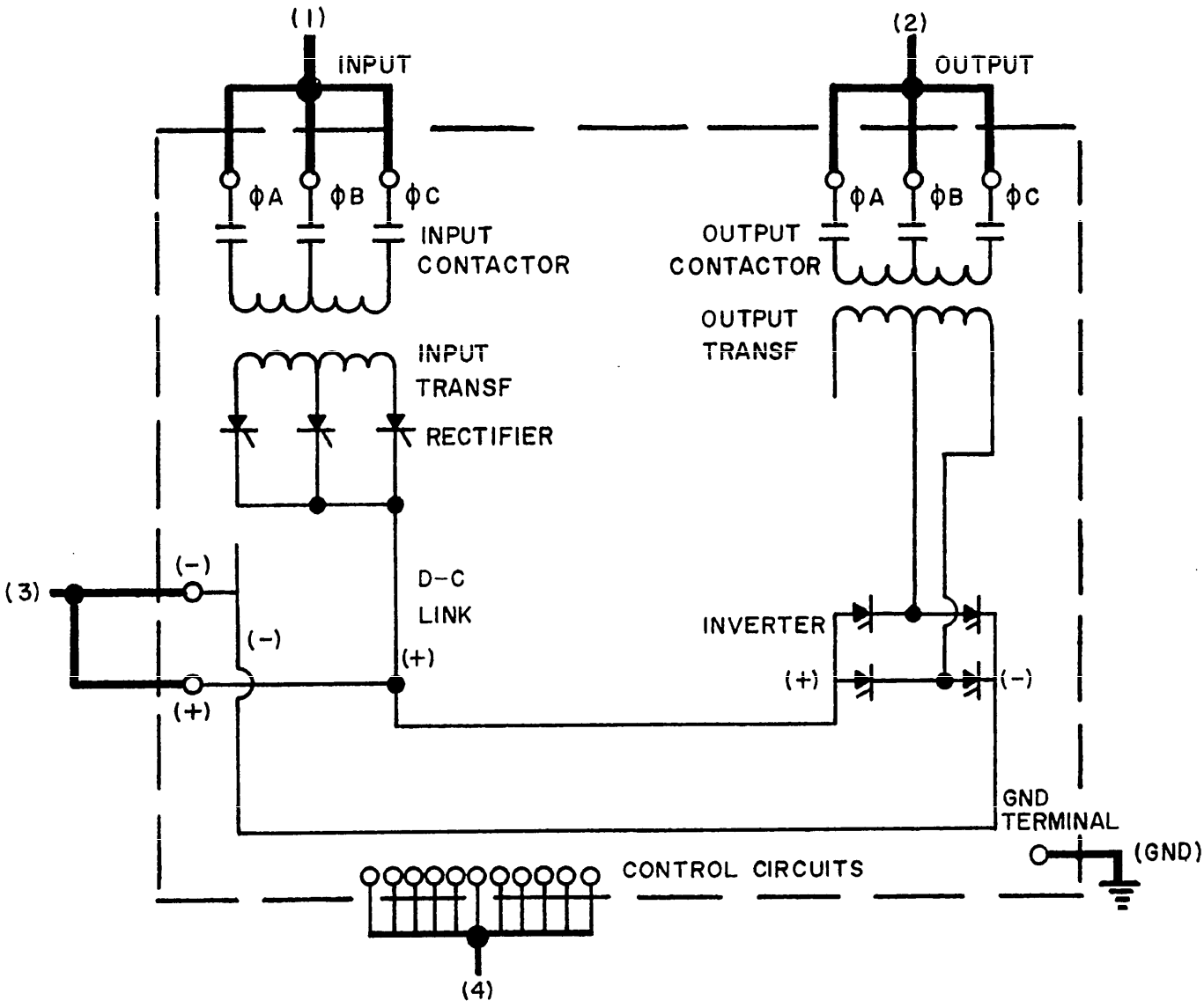
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SH 12427

Figure 7. Structureborne noise acceptance levels.

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SH 12428

FIGURE 8. Circuit connections for insulation resistance and dielectric withstanding voltage test.

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Notes to figure 8:

1. Short the three input terminals (phases A, B, and C) together and form a common terminal labeled (1).

2. Short the three output terminals (phases A, B, and C) together and form a common terminal labeled (2).

3. Short the positive and negative sides of the DC link to a common terminal labeled (3).

4. Remove the control drawer from the unit. Insert four special connectors with all pins shorted together into the cabinet control wiring connectors. Terminate all of the shorted pins of the four special connectors to a common terminal labeled (4).

5. Connect the unit ground terminal to ground and label (GND).

6. Place shorting jumpers over each contact of the input and output contactors. Place shorting jumpers over each power silicon control rectifier (SCR) in the rectifier drawers and in the switching drawers (inverter).

7. Number of measurements are:

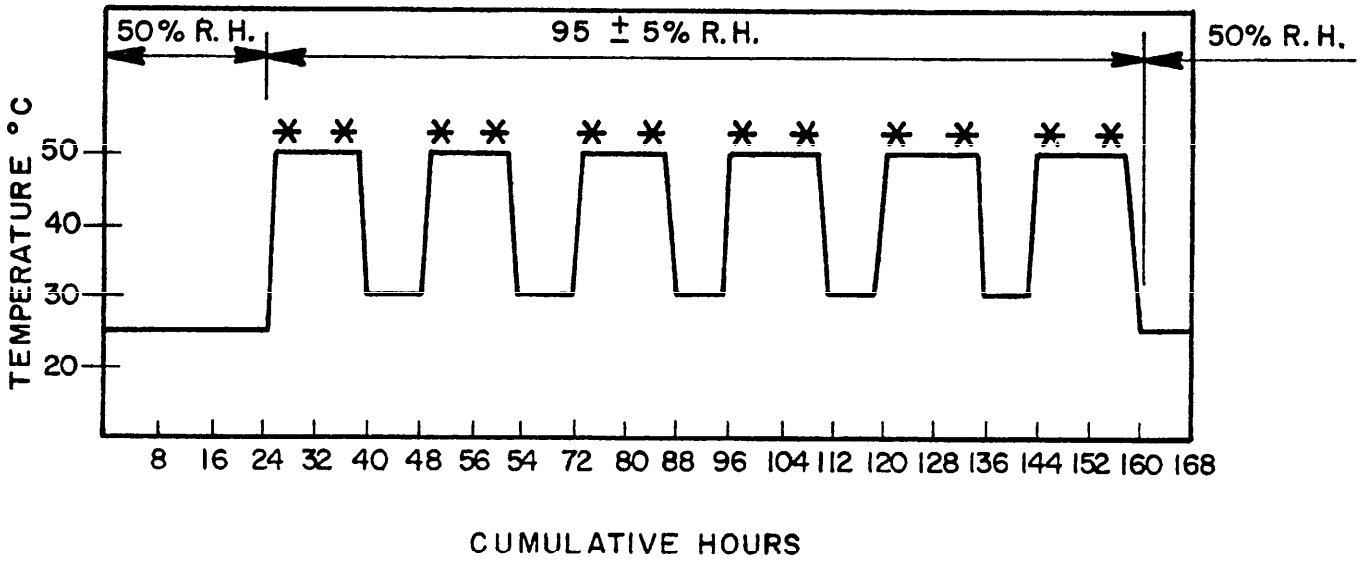
   (a) (1) to (2), (3), (4) and (GND) connected.

   (b) (2) to (1), (3), (4) and (GND) connected.

   (c) (3) to (1), (2), (4) and (GND) connected.

   (d) (4) to (1), (2), (3) and (GND) connected.

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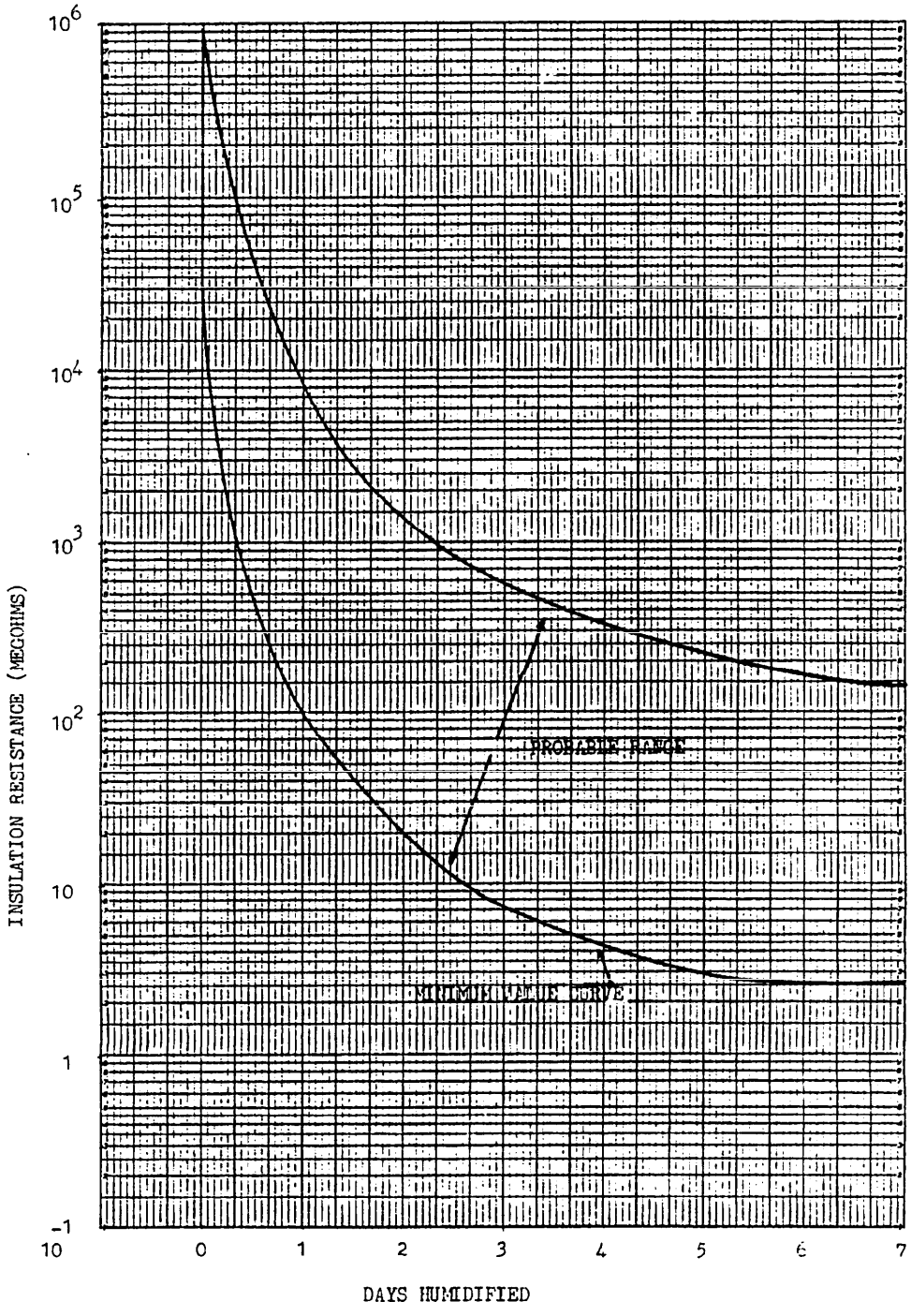


* Indicates approximate time measurements are to be made.

SH 12429

FIGURE 9. Humidification cycle.

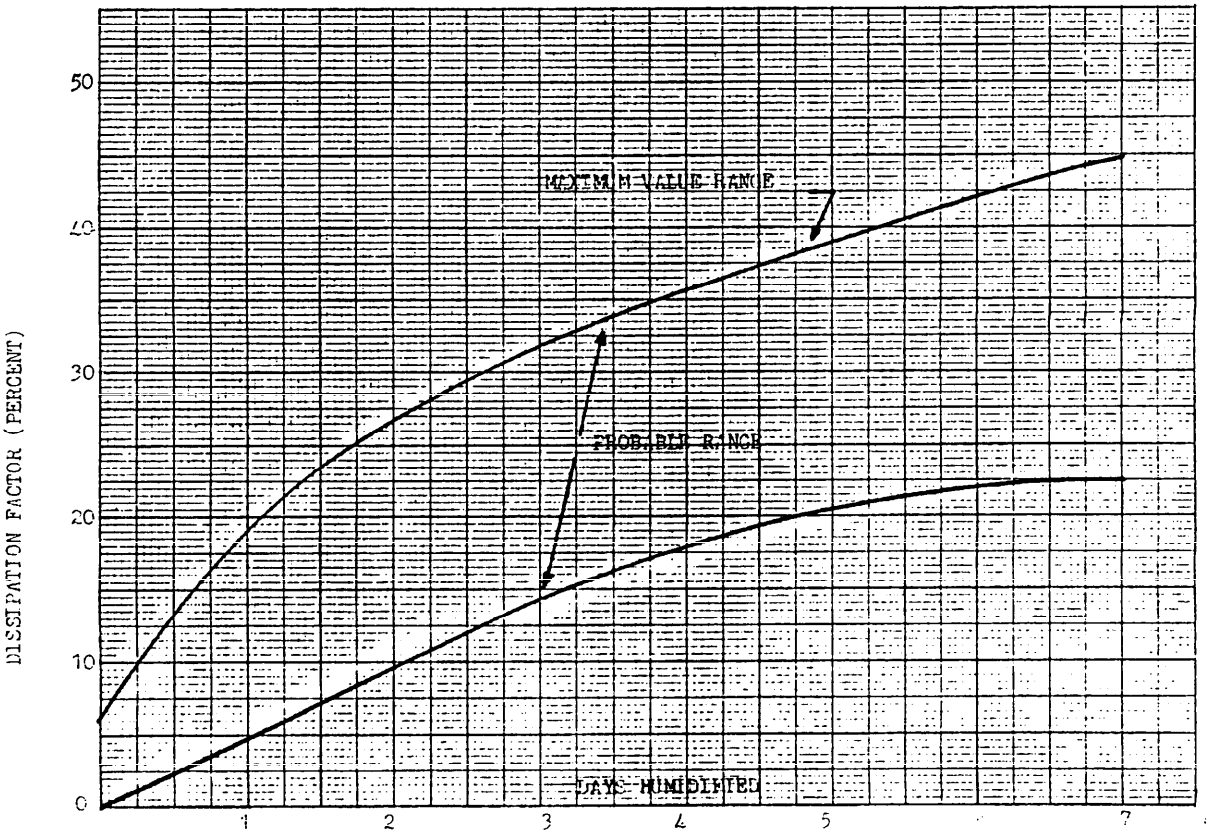
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SH 11293

FIGURE 10. Insulation resistance.

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SH 11294

FIGURE 11. Dissipation factor.

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or grounding or parts or circuit elements during the test to prevent by-passing

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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MIL-F-24122C(SH)	Frequency Changer, Solid State (Naval Shipboard)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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